Hartree-Fock Theory

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Szabo and Ostlund's book

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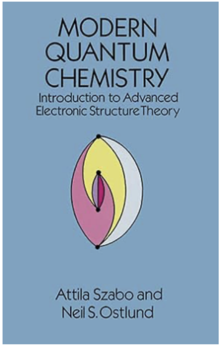

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by Attila Szabo (Author), Neil S. Ostlund (Author) | Format: Paperback

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The aim of this graduate-level textbook is to present and explain, at other than a superficial level, modem *ab initio* approaches to the calculation of the electronic structure and properties of molecules. The first three chapters contain introductory material culminating in a thorough discussion of the Hartree-Fock approximation.The remaining four chapters describe a variety of more sophisticated approaches, which improve upon this approximation.

Among the highlights of the seven chapters are (1) a review of the mathematics (mostly matrix algebra) required for the rest of the book, (2) an introduction to the basic techniques, ideas, and notations of quantum chemistry, (3) a thorough discussion of the Hartree-Fock approximation, (4) a treatment of configuration interaction (CI) and approaches incorporating electron correlation, (5) a description of the independent electron pair approximation and a variety of more sophisticated approaches that incorporate coupling between pairs, (6) a consideration of the perturbative approach to the calculation of the correlation energy of many-electron systems and (7) a brief introduction to the use of the one-particle many-body Green's function in quantum chemistry.

Over 150 exercises, designed to help the reader acquire a working knowledge of the material, are embedded in the text. The book is largely self-contained and requires no prerequisite other than a solid undergraduate physical chemistry course;

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Outline

Foundations

1. **The electronic problem**
   Molecular Hamiltonian, Born-Oppenheimer approximation, and Pauli principle

2. **Hartree-Fock approximation**
   Slater determinants, Hartree–Fock energy, Coulomb and exchange operators

3. **Determinants and spin**
   Excited determinants, Slater-Condon rules, spin adaptation and spin contamination

Equations and solutions

4. **Finite-basis Hartree-Fock**
   Fock operator, orbital energies, Roothaan–Hall equations and Koopmans' theorem

5. **Hartree-Fock in practice**
   SCF convergence, DIIS, excited-state SCF, properties and orbital localization

6. **Beyond closed-shell RHF**
   UHF, symmetry breaking in H_2, the Coulson–Fischer point and GHF

Motivations & Assumptions

- We consider the time-independent Schrödinger equation
- Hartree-Fock (HF) is an ab initio method: it contains no empirical parameters, apart from the choice of basis set
- We neglect relativistic effects
- HF is a mean-field approximation, i.e., interactions are taken into account in an average fashion
- HF is the starting point for much of electronic structure theory

The Hamiltonian

In the Schrödinger equation

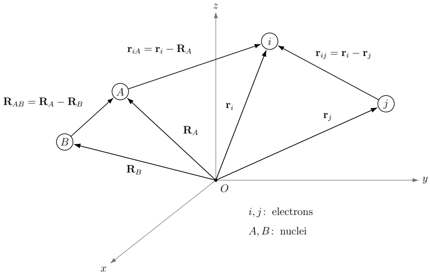
$$\hat{\mathcal{H}}\Phi(\{\boldsymbol{r}_i\}, \{\boldsymbol{R}_A\}) = \mathcal{E}\Phi(\{\boldsymbol{r}_i\}, \{\boldsymbol{R}_A\})$$

the total Hamiltonian is

$$\boxed{\hat{\mathcal{H}} = \hat{\mathcal{T}}_{\mathrm{n}} + \hat{\mathcal{T}}_{\mathrm{e}} + \hat{\mathcal{V}}_{\mathrm{ne}} + \hat{\mathcal{V}}_{\mathrm{ee}} + \hat{\mathcal{V}}_{\mathrm{nn}}}$$

What are all these terms?

- $\hat{\mathcal{T}}_{\mathrm{n}}$ is the kinetic energy of the nuclei
- $\hat{\mathcal{T}}_{\mathrm{e}}$ is the kinetic energy of the electrons
- $\hat{\mathcal{V}}_{\mathrm{ne}}$ is the Coulomb attraction between nuclei and electrons
- $\hat{\mathcal{V}}_{\mathrm{ee}}$ is the Coulomb repulsion between electrons
- $\hat{\mathcal{V}}_{\mathrm{nn}}$ is the Coulomb repulsion between nuclei



Hamiltonian terms in atomic units

In atomic units ($m = e = \hbar = 1$)

$$\hat{\mathcal{T}}_{\text{n}} = -\sum_{A=1}^{M} \frac{\nabla_A^2}{2M_A}$$

$$\hat{\mathcal{T}}_{\text{e}} = -\sum_{i=1}^{N} \frac{\nabla_i^2}{2}$$

$$\hat{\mathcal{V}}_{\text{ne}} = -\sum_{A=1}^{M} \sum_{i=1}^{N} \frac{Z_A}{r_{iA}}$$

$$\hat{\mathcal{V}}_{\text{ee}} = \sum_{i<j}^{N} \frac{1}{r_{ij}}$$

$$\hat{\mathcal{V}}_{\text{nn}} = \sum_{A<B}^{M} \frac{Z_A Z_B}{R_{AB}}$$

- ∇^2 is the Laplace operator (or Laplacian)
- M_A is the mass of nucleus A
- Z_A is the charge of nucleus A
- r_{iA} is the distance between electron i and nucleus A
- r_{ij} is the distance between electrons i and j
- R_{AB} is the distance between nuclei A and B

The Born-Oppenheimer approximation

Born-Oppenheimer approximation = decoupling nuclei and electrons

Because $M_A \gg 1$, the electrons in a molecule move in the field of fixed nuclei

$$\Phi(\{\boldsymbol{r}_i\}, \{\boldsymbol{R}_A\}) = \Phi_{\text{nucl}}(\{\boldsymbol{R}_A\})\Phi_{\text{elec}}(\{\boldsymbol{r}_i\}, \{\boldsymbol{R}_A\}) \quad \text{with} \quad \mathcal{E}_{\text{tot}} = \mathcal{E}_{\text{elec}} + \sum_{A<B}^{M} \frac{Z_A Z_B}{R_{AB}}$$

Electronic Hamiltonian

The electronic Hamiltonian is

$$\hat{\mathcal{H}}_{\text{elec}}\Phi_{\text{elec}} = \mathcal{E}_{\text{elec}}\Phi_{\text{elec}} \quad \text{with} \quad \boxed{\hat{\mathcal{H}}_{\text{elec}} = \hat{\mathcal{T}}_{\text{e}} + \hat{\mathcal{V}}_{\text{ne}} + \hat{\mathcal{V}}_{\text{ee}}}$$

Nuclear Hamiltonian

The nuclear Hamiltonian is

$$\hat{\mathcal{H}}_{\text{nucl}}\Phi_{\text{nucl}} = \mathcal{E}_{\text{nucl}}\Phi_{\text{nucl}} \quad \text{with} \quad \boxed{\hat{\mathcal{H}}_{\text{nucl}} = \hat{\mathcal{T}}_{\text{n}} + \hat{\mathcal{V}}_{\text{nn}} + \mathcal{E}_{\text{elec}}(\{\boldsymbol{R}_A\})}$$

It describes nuclear motions on the potential energy surface characterized by $\hat{\mathcal{V}}_{\text{nn}} + \mathcal{E}_{\text{elec}}(\{\boldsymbol{R}_A\})$

Spin of the electron

We are interested in electrons, which are fermions $\Rightarrow$ Pauli exclusion principle (see next slide)

Spin functions: $|\omega\rangle = |s, m_s\rangle$ $\quad s^2 |s, m_s\rangle = s(s+1) |s, m_s\rangle$ $\quad s_z |s, m_s\rangle = m_s |s, m_s\rangle$

$|\alpha\rangle = \left|\frac{1}{2}, +\frac{1}{2}\right\rangle$ = spin-up electron $\qquad |\beta\rangle = \left|\frac{1}{2}, -\frac{1}{2}\right\rangle$ = spin-down electron

$$\int \alpha^*(\omega)\beta(\omega)\, d\omega = \int \beta^*(\omega)\alpha(\omega)\, d\omega = 0 \qquad \int \alpha^*(\omega)\alpha(\omega)\, d\omega = \int \beta^*(\omega)\beta(\omega)\, d\omega = 1$$

$$\langle\alpha|\beta\rangle = \langle\beta|\alpha\rangle = 0 \qquad \langle\alpha|\alpha\rangle = \langle\beta|\beta\rangle = 1$$

The composite variable $\boldsymbol{x}$ combines spin (ω) and spatial ($\boldsymbol{r}$) coordinates: $\boxed{\boldsymbol{x} = (\omega, \boldsymbol{r})}$

Pauli exclusion principle

$$\hat{\mathcal{H}}_{\text{elec}}\Phi(\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N) = \mathcal{E}_{\text{elec}}\Phi(\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N)$$

$$\Phi(\boldsymbol{x}_1, \ldots, \boldsymbol{x}_i, \ldots, \boldsymbol{x}_j, \ldots, \boldsymbol{x}_N) = -\Phi(\boldsymbol{x}_1, \ldots, \boldsymbol{x}_j, \ldots, \boldsymbol{x}_i, \ldots, \boldsymbol{x}_N)$$

Bosons vs Fermions

Symmetry vs Antisymmetry

Indistinguishable particles mean

$$\boxed{|\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2)|^2 = |\Psi(\boldsymbol{x}_2, \boldsymbol{x}_1)|^2 \quad \Rightarrow \quad \Psi(\boldsymbol{x}_2, \boldsymbol{x}_1) = e^{i\theta}\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2)}$$

Applying twice the permutation operator $\hat{\mathcal{P}}_{12}$ gives

$$\hat{\mathcal{P}}_{12}\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2) = \Psi(\boldsymbol{x}_2, \boldsymbol{x}_1) = e^{i\theta}\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2) \qquad \hat{\mathcal{P}}_{12}^2\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2) = e^{2i\theta}\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2) = \Psi(\boldsymbol{x}_1, \boldsymbol{x}_2)$$

$$\Rightarrow \quad e^{2i\theta} = 1 \quad \Rightarrow \quad \theta = n\pi \quad \Rightarrow \quad \boxed{e^{i\theta} = \pm 1}$$

Bosons mean $\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2) = \Psi(\boldsymbol{x}_2, \boldsymbol{x}_1)$ and Fermions mean $\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2) = -\Psi(\boldsymbol{x}_2, \boldsymbol{x}_1)$

Let's put them at the same spot, i.e. $\boldsymbol{x} = \boldsymbol{x}_1 = \boldsymbol{x}_2$

For Fermions, $\Psi(\boldsymbol{x}, \boldsymbol{x}) = -\Psi(\boldsymbol{x}, \boldsymbol{x}) \quad \Rightarrow \quad \boxed{\Psi(\boldsymbol{x}, \boldsymbol{x}) = 0}$

The wavefunction vanishes: this is the origin of the Fermi (exchange) hole

Antisymmetry

Problem:

"Given two one-electron functions $\chi_1(\boldsymbol{x})$ and $\chi_2(\boldsymbol{x})$, could you construct a two-electron (fermionic) wavefunction $\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2)$?"

Antisymmetry

Problem:

"Given two one-electron functions $\chi_1(\boldsymbol{x})$ and $\chi_2(\boldsymbol{x})$, could you construct a two-electron (fermionic) wavefunction $\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2)$?"

Solution

A possible solution is

$$\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2) = \frac{1}{\sqrt{2}}[\chi_1(\boldsymbol{x}_1)\chi_2(\boldsymbol{x}_2) - \chi_2(\boldsymbol{x}_1)\chi_1(\boldsymbol{x}_2)]$$

This has been popularized by Slater:

$$\boxed{\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2) = \frac{1}{\sqrt{2}}\begin{vmatrix}\chi_1(\boldsymbol{x}_1) & \chi_2(\boldsymbol{x}_1)\\ \chi_1(\boldsymbol{x}_2) & \chi_2(\boldsymbol{x}_2)\end{vmatrix} = \frac{1}{\sqrt{2}}[\chi_1(\boldsymbol{x}_1)\chi_2(\boldsymbol{x}_2) - \chi_2(\boldsymbol{x}_1)\chi_1(\boldsymbol{x}_2)]}$$

This is called a Slater determinant!

A wavefunction of the form $\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2) = \chi_1(\boldsymbol{x}_1)\chi_2(\boldsymbol{x}_2)$ is called a Hartree product

The Hartree-Fock wavefunction

A Slater determinant

$$\Psi_{\mathrm{HF}}(\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N) = \frac{1}{\sqrt{N!}} \begin{vmatrix} \chi_1(\boldsymbol{x}_1) & \chi_2(\boldsymbol{x}_1) & \cdots & \chi_N(\boldsymbol{x}_1) \\ \chi_1(\boldsymbol{x}_2) & \chi_2(\boldsymbol{x}_2) & \cdots & \chi_N(\boldsymbol{x}_2) \\ \vdots & \vdots & \ddots & \vdots \\ \chi_1(\boldsymbol{x}_N) & \chi_2(\boldsymbol{x}_N) & \cdots & \chi_N(\boldsymbol{x}_N) \end{vmatrix} \equiv |\chi_1(\boldsymbol{x}_1)\chi_2(\boldsymbol{x}_2)\ldots\chi_N(\boldsymbol{x}_N)\rangle$$

$$= \hat{\mathcal{A}}[\chi_1(\boldsymbol{x}_1)\chi_2(\boldsymbol{x}_2)\ldots\chi_N(\boldsymbol{x}_N)] = \hat{\mathcal{A}}\Pi(\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N)$$

- $\hat{\mathcal{A}}$ is called the antisymmetrizer
- $\Pi(\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N)$ is a Hartree product
- The many-electron wavefunction $\Psi_{\mathrm{HF}}(\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N)$ is an antisymmetrized product of one-electron functions

Spin orbitals

$$\chi_p(\boldsymbol{x}) = \underbrace{\sigma(\omega)}_{\text{spin}}\underbrace{\psi_p^\sigma(\boldsymbol{r})}_{\text{spatial}} \quad \text{with} \quad \sigma(\omega) = \begin{cases} \alpha(\omega) \\ \beta(\omega) \end{cases} \quad \text{and} \quad \psi_p^\sigma(\boldsymbol{r}) = \underbrace{\sum_\mu^K C_{\mu p}^\sigma \phi_\mu(\boldsymbol{r})}_{\text{LCAO}}$$

The spin orbitals are orthogonal

$$\begin{aligned} \langle \chi_p | \chi_q \rangle &= \int \chi_p^*(\boldsymbol{x}) \chi_q(\boldsymbol{x}) \, \mathrm{d}\boldsymbol{x} \\ &= \delta_{pq} = \begin{cases} 1 & \text{if } p = q \\ 0 & \text{otherwise} \end{cases} \quad = \text{Kronecker delta} \end{aligned}$$

The basis functions are, a priori, not orthogonal

$$\begin{aligned} \langle \phi_\mu | \phi_\nu \rangle &= \int \phi_\mu^*(\boldsymbol{r}) \phi_\nu(\boldsymbol{r}) \, \mathrm{d}\boldsymbol{r} \\ &= S_{\mu\nu} \quad = \text{Overlap matrix elements} \end{aligned}$$

Restricted vs Unrestricted Hartree-Fock

Restricted formalism

- Restricted HF (RHF) is made to describe closed-shell systems

$$\chi_p^{\text{RHF}}(\boldsymbol{x}) = \begin{cases} \alpha(\omega)\psi_p(\boldsymbol{r}) \\ \beta(\omega)\psi_p(\boldsymbol{r}) \end{cases} = \begin{cases} \psi_p(\boldsymbol{r}) \\ \bar{\psi}_p(\boldsymbol{r}) \end{cases}$$

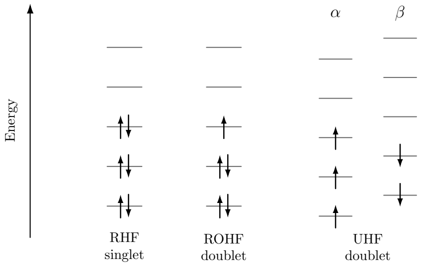
- The closed-shell RHF ansatz does **not** describe open-shell systems

Unrestricted formalism

- For open-shell systems, one can use unrestricted HF (UHF)

$$\chi_p^{\text{UHF}}(\boldsymbol{x}) = \begin{cases} \alpha(\omega)\psi_p^{\alpha}(\boldsymbol{r}) \\ \beta(\omega)\psi_p^{\beta}(\boldsymbol{r}) \end{cases}$$

- UHF may break spin symmetry!
- Restricted open-shell HF (ROHF) retains shared spatial orbitals (will not be discussed here!)



Spatial orbitals

The restricted spatial orbitals are orthogonal

$$\langle \psi_p | \psi_q \rangle = \int \psi_p^*(\boldsymbol{r}) \psi_q(\boldsymbol{r}) \, \mathrm{d}\boldsymbol{r} = \delta_{pq}$$

The same-spin unrestricted spatial orbitals are orthogonal

$$\left\langle \psi_p^{\alpha} \middle| \psi_q^{\alpha} \right\rangle = \int \psi_p^{\alpha *}(\boldsymbol{r}) \psi_q^{\alpha}(\boldsymbol{r}) \, \mathrm{d}\boldsymbol{r} = \delta_{pq}$$

$$\left\langle \psi_p^{\beta} \middle| \psi_q^{\beta} \right\rangle = \int \psi_p^{\beta *}(\boldsymbol{r}) \psi_q^{\beta}(\boldsymbol{r}) \, \mathrm{d}\boldsymbol{r} = \delta_{pq}$$

The opposite-spin unrestricted spatial orbitals are, a priori, not orthogonal

$$\left\langle \psi_p^{\alpha} \middle| \psi_q^{\beta} \right\rangle = \int \psi_p^{\alpha *}(\boldsymbol{r}) \psi_q^{\beta}(\boldsymbol{r}) \, \mathrm{d}\boldsymbol{r} = S_{pq}^{\alpha\beta}$$

Orbital spaces and index conventions

Orbitals

- $\{\phi_\mu | \mu = 1, \ldots, K\}$ are basis functions or atomic orbitals (AOs)
- $\{\psi_p | p = 1, \ldots, K\}$ are (restricted) spatial orbitals or molecular orbitals (MOs)
- $\{\chi_p | p = 1, \ldots, 2K\}$ are spin orbitals

Indices

With K basis functions, one can create

- K spatial orbitals and $2K$ spin orbitals $\Rightarrow$ indices $p, q, r, s, \ldots$
- N spin orbitals are occupied $\Rightarrow$ indices $i, j, k, l, \ldots$
- $2K - N$ spin orbitals are unoccupied/vacant $\Rightarrow$ indices $a, b, c, d, \ldots$

Closed vs Open

- When a system has 2 electrons in each orbital, it is called a closed-shell system
- Otherwise, it is called an open-shell system
- For the ground state of a closed shell, the first $N/2$ spatial orbitals are doubly-occupied and the last $K - N/2$ are vacant

HF energy

$$\boxed{E_{\text{HF}} = \langle \Psi_{\text{HF}} | \hat{\mathcal{H}}_{\text{elec}} | \Psi_{\text{HF}} \rangle} \quad \text{where} \quad \hat{\mathcal{H}}_{\text{elec}} = \hat{\mathcal{T}}_{\text{e}} + \hat{\mathcal{V}}_{\text{ne}} + \hat{\mathcal{V}}_{\text{ee}}$$

One-electron Hamiltonian (or core Hamiltonian)

$$\hat{O}_1 = \hat{\mathcal{T}}_{\text{e}} + \hat{\mathcal{V}}_{\text{ne}} = \sum_{i=1}^{N} \hat{h}(\boldsymbol{x}_i) \quad \text{where} \quad \hat{h}(\boldsymbol{x}_i) = -\frac{\nabla_i^2}{2} - \sum_{A=1}^{M} \frac{Z_A}{r_{iA}}$$

Two-electron Hamiltonian (electron-electron repulsion)

$$\hat{O}_2 = \hat{\mathcal{V}}_{\text{ee}} = \sum_{i<j}^{N} \frac{1}{r_{ij}}$$

One-electron contributions to the HF energy

Electronic Hamiltonian

$$\boxed{\hat{\mathcal{H}}_{\text{elec}} = \hat{O}_1 + \hat{O}_2 = \sum_{i=1}^{N} \hat{h}(\boldsymbol{x}_i) + \sum_{i<j}^{N} \frac{1}{r_{ij}}}$$

Nuclear repulsion

$$\langle \Psi_{\text{HF}} | \hat{\mathcal{V}}_{\text{nn}} | \Psi_{\text{HF}} \rangle = V_{\text{nn}} \langle \Psi_{\text{HF}} | \Psi_{\text{HF}} \rangle = V_{\text{nn}} \quad \text{if} \quad \langle \Psi_{\text{HF}} | \Psi_{\text{HF}} \rangle = 1$$

Core Hamiltonian

$$\langle \Psi_{\text{HF}} | \hat{O}_1 | \Psi_{\text{HF}} \rangle = \sum_{i=1}^{N} \langle \chi_i(\boldsymbol{x}_1) | \hat{h}(\boldsymbol{x}_1) | \chi_i(\boldsymbol{x}_1) \rangle = \sum_{i=1}^{N} h_{ii}$$

Two-electron contributions to the HF energy

Two-electron Hamiltonian

$$\langle \Psi_{\mathrm{HF}} | \hat{O}_2 | \Psi_{\mathrm{HF}} \rangle = \sum_{i<j}^{N} \Big[\big\langle \chi_i(\boldsymbol{x}_1)\chi_j(\boldsymbol{x}_2) \big| r_{12}^{-1} \big| \chi_i(\boldsymbol{x}_1)\chi_j(\boldsymbol{x}_2) \big\rangle - \big\langle \chi_i(\boldsymbol{x}_1)\chi_j(\boldsymbol{x}_2) \big| r_{12}^{-1} \big| \chi_j(\boldsymbol{x}_1)\chi_i(\boldsymbol{x}_2) \big\rangle \Big]$$

$$= \sum_{i<j}^{N} \Bigg(\underbrace{\mathcal{J}_{ij}}_{\text{Coulomb}} - \underbrace{\mathcal{K}_{ij}}_{\text{Exchange}} \Bigg) = \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} \left(\mathcal{J}_{ij} - \mathcal{K}_{ij} \right) \quad \text{because} \quad \boxed{\mathcal{J}_{ii} = \mathcal{K}_{ii}}$$

HF energy

$$\boxed{E_{\mathrm{HF}} = \sum_{i=1}^{N} h_{ii} + \sum_{i<j}^{N} \left(\mathcal{J}_{ij} - \mathcal{K}_{ij} \right)}$$

Coulomb operator

Coulomb operator

$$\hat{\mathcal{J}}_j(\boldsymbol{x}_1)\,|\chi_i(\boldsymbol{x}_1)\rangle = \left\langle \chi_j(\boldsymbol{x}_2) \middle| r_{12}^{-1} \middle| \chi_j(\boldsymbol{x}_2) \right\rangle |\chi_i(\boldsymbol{x}_1)\rangle = \left[\int \chi_j^*(\boldsymbol{x}_2) r_{12}^{-1} \chi_j(\boldsymbol{x}_2)\,\mathrm{d}\boldsymbol{x}_2 \right] |\chi_i(\boldsymbol{x}_1)\rangle$$

Coulomb matrix elements

$$\begin{aligned}\mathcal{J}_{ij} &= \langle \chi_i(\boldsymbol{x}_1) | \hat{\mathcal{J}}_j(\boldsymbol{x}_1) | \chi_i(\boldsymbol{x}_1) \rangle \\ &= \left\langle \chi_i(\boldsymbol{x}_1)\chi_j(\boldsymbol{x}_2) \middle| r_{12}^{-1} \middle| \chi_i(\boldsymbol{x}_1)\chi_j(\boldsymbol{x}_2) \right\rangle \\ &= \iint \chi_i^*(\boldsymbol{x}_1)\chi_j^*(\boldsymbol{x}_2) r_{12}^{-1} \chi_i(\boldsymbol{x}_1)\chi_j(\boldsymbol{x}_2)\,\mathrm{d}\boldsymbol{x}_1\,\mathrm{d}\boldsymbol{x}_2\end{aligned}$$

Exchange operator

(non-local) Exchange operator

$$\begin{aligned}\hat{\mathcal{K}}_j(\boldsymbol{x}_1)\,|\chi_i(\boldsymbol{x}_1)\rangle &= \left\langle \chi_j(\boldsymbol{x}_2)\middle|r_{12}^{-1}\middle|\chi_i(\boldsymbol{x}_2)\right\rangle |\chi_j(\boldsymbol{x}_1)\rangle \\ &= \left[\int \chi_j^*(\boldsymbol{x}_2) r_{12}^{-1} \chi_i(\boldsymbol{x}_2)\,\mathrm{d}\boldsymbol{x}_2\right] |\chi_j(\boldsymbol{x}_1)\rangle\end{aligned}$$

Exchange matrix elements

$$\begin{aligned}\mathcal{K}_{ij} &= \langle\chi_i(\boldsymbol{x}_1)|\hat{\mathcal{K}}_j(\boldsymbol{x}_1)|\chi_i(\boldsymbol{x}_1)\rangle \\ &= \left\langle \chi_i(\boldsymbol{x}_1)\chi_j(\boldsymbol{x}_2)\middle|r_{12}^{-1}\middle|\chi_j(\boldsymbol{x}_1)\chi_i(\boldsymbol{x}_2)\right\rangle \\ &= \iint \chi_i^*(\boldsymbol{x}_1)\chi_j^*(\boldsymbol{x}_2) r_{12}^{-1} \chi_j(\boldsymbol{x}_1)\chi_i(\boldsymbol{x}_2)\,\mathrm{d}\boldsymbol{x}_1\,\mathrm{d}\boldsymbol{x}_2\end{aligned}$$

Notations for one-electron integrals and electron repulsion integrals (ERIs)

Dirac or Physicists' notations

$$\langle i|\hat{h}|j\rangle = h_{ij} = \left\langle \chi_i \middle| \hat{h} \middle| \chi_j \right\rangle = \int \chi_i^*(\boldsymbol{x}_1)\hat{h}(\boldsymbol{x}_1)\chi_j(\boldsymbol{x}_1)\,\mathrm{d}\boldsymbol{x}_1$$

$$\langle ij|kl\rangle = \left\langle \chi_i\chi_j \middle| \chi_k\chi_l \right\rangle = \iint \chi_i^*(\boldsymbol{x}_1)\chi_j^*(\boldsymbol{x}_2)\frac{1}{r_{12}}\chi_k(\boldsymbol{x}_1)\chi_l(\boldsymbol{x}_2)\,\mathrm{d}\boldsymbol{x}_1\,\mathrm{d}\boldsymbol{x}_2$$

$$\langle ij||kl\rangle = \langle ij|kl\rangle - \langle ij|lk\rangle = \iint \chi_i^*(\boldsymbol{x}_1)\chi_j^*(\boldsymbol{x}_2)\frac{1}{r_{12}}\left(1 - \hat{\mathcal{P}}_{12}\right)\chi_k(\boldsymbol{x}_1)\chi_l(\boldsymbol{x}_2)\,\mathrm{d}\boldsymbol{x}_1\,\mathrm{d}\boldsymbol{x}_2$$

Mulliken or Chemists' notations

$$(i|\hat{h}|j) = h_{ij} = (\chi_i|\hat{h}|\chi_j) = \int \chi_i^*(\boldsymbol{x}_1)\hat{h}(\boldsymbol{x}_1)\chi_j(\boldsymbol{x}_1)\,\mathrm{d}\boldsymbol{x}_1$$

$$(ij|kl) = (\chi_i\chi_j|\chi_k\chi_l) = \iint \chi_i^*(\boldsymbol{x}_1)\chi_j(\boldsymbol{x}_1)\frac{1}{r_{12}}\chi_k^*(\boldsymbol{x}_2)\chi_l(\boldsymbol{x}_2)\,\mathrm{d}\boldsymbol{x}_1\,\mathrm{d}\boldsymbol{x}_2$$

Warning! We will use the same notation for spin and spatial orbitals!

Permutation symmetry

Permutation symmetry in physicists' notation

$$\langle ij|kl\rangle = \left\langle \chi_i\chi_j \middle| \chi_k\chi_l \right\rangle = \iint \chi_i^*(\boldsymbol{x}_1)\chi_j^*(\boldsymbol{x}_2)\frac{1}{r_{12}}\chi_k(\boldsymbol{x}_1)\chi_l(\boldsymbol{x}_2)\,\mathrm{d}\boldsymbol{x}_1\,\mathrm{d}\boldsymbol{x}_2$$

Complex-valued integrals: $\langle ij|kl\rangle = \langle ji|lk\rangle = \langle kl|ij\rangle^* = \langle lk|ji\rangle^*$

Real-valued integrals: $\langle ij|kl\rangle = \langle ji|lk\rangle = \langle kl|ij\rangle = \langle lk|ji\rangle = \langle kj|il\rangle = \langle jk|li\rangle = \langle il|kj\rangle = \langle li|jk\rangle$

Permutation symmetry in chemists' notation

$$(ij|kl) = (\chi_i\chi_j|\chi_k\chi_l) = \iint \chi_i^*(\boldsymbol{x}_1)\chi_j(\boldsymbol{x}_1)\frac{1}{r_{12}}\chi_k^*(\boldsymbol{x}_2)\chi_l(\boldsymbol{x}_2)\,\mathrm{d}\boldsymbol{x}_1\,\mathrm{d}\boldsymbol{x}_2$$

Real-valued integrals: $(ij|kl) = (ji|kl) = (ij|lk) = (ji|lk) = (kl|ij) = (lk|ij) = (kl|ji) = (lk|ji)$

For real-valued orbitals, the ERIs possess 8-fold permutation symmetry!

Normalization of a two-electron determinant

Problem: Normalization of the HF wavefunction

"Show that the HF wavefunction built with two (normalized) orthogonal spin orbitals χ_1 *and* χ_2 *is normalized"*

Normalization of a two-electron determinant

Problem: Normalization of the HF wavefunction

"Show that the HF wavefunction built with two (normalized) orthogonal spin orbitals χ_1 and χ_2 is normalized"

Solution

$$\Psi_{\text{HF}} = \frac{1}{\sqrt{2}} \begin{vmatrix} \chi_1(1) & \chi_2(1) \\ \chi_1(2) & \chi_2(2) \end{vmatrix} = \frac{\chi_1(1)\chi_2(2) - \chi_1(2)\chi_2(1)}{\sqrt{2}}$$

$$\begin{aligned}
\langle \Psi_{\text{HF}} | \Psi_{\text{HF}} \rangle &= \frac{1}{2} \langle \chi_1(1)\chi_2(2) - \chi_2(1)\chi_1(2) | \chi_1(1)\chi_2(2) - \chi_2(1)\chi_1(2) \rangle \\
&= \frac{1}{2} \Big[\langle \chi_1(1)\chi_2(2) | \chi_1(1)\chi_2(2) \rangle - \langle \chi_1(1)\chi_2(2) | \chi_2(1)\chi_1(2) \rangle \\
&\quad - \langle \chi_2(1)\chi_1(2) | \chi_1(1)\chi_2(2) \rangle + \langle \chi_2(1)\chi_1(2) | \chi_2(1)\chi_1(2) \rangle \Big] \\
&= \frac{1}{2} \Big[1 - 0 - 0 + 1 \Big] = 1
\end{aligned}$$

Remember that $\langle \chi_1(1)\chi_2(2) | \chi_1(1)\chi_2(2) \rangle = \langle \chi_1(1) | \chi_1(1) \rangle \langle \chi_2(2) | \chi_2(2) \rangle$

One-electron energy of a two-electron determinant

Problem: Core Hamiltonian

"Show that $\langle \Psi_{HF} | \hat{O}_1 | \Psi_{HF} \rangle = \sum_{i=1}^{N} h_{ii}$ for the same system"

One-electron energy of a two-electron determinant

Problem: Core Hamiltonian

"Show that $\langle\Psi_{HF}|\hat{O}_1|\Psi_{HF}\rangle = \sum_{i=1}^{N} h_{ii}$ for the same system"

Solution

$$\hat{O}_1 = \hat{h}(1) + \hat{h}(2)$$

$$\begin{aligned}
&\langle\Psi_{\mathrm{HF}}|\hat{h}(1) + \hat{h}(2)|\Psi_{\mathrm{HF}}\rangle \\
&\quad= \frac{1}{2}\langle\chi_1(1)\chi_2(2) - \chi_1(2)\chi_2(1)|\hat{h}(1) + \hat{h}(2)|\chi_1(1)\chi_2(2) - \chi_1(2)\chi_2(1)\rangle \\
&\quad= \frac{1}{2}\Big[\langle\chi_1(1)\chi_2(2)|\hat{h}(1) + \hat{h}(2)|\chi_1(1)\chi_2(2)\rangle - \langle\chi_1(1)\chi_2(2)|\hat{h}(1) + \hat{h}(2)|\chi_2(1)\chi_1(2)\rangle \\
&\quad- \langle\chi_2(1)\chi_1(2)|\hat{h}(1) + \hat{h}(2)|\chi_1(1)\chi_2(2)\rangle + \langle\chi_2(1)\chi_1(2)|\hat{h}(1) + \hat{h}(2)|\chi_2(1)\chi_1(2)\rangle\Big] \\
&\quad= \frac{1}{2}\Big[h_{11} + h_{22} - 0 - 0 + h_{22} + h_{11}\Big] = h_{11} + h_{22}
\end{aligned}$$

Electron interaction in a two-electron determinant

Problem: Two-electron Hamiltonian

"Show that $\langle \Psi_{HF} | \hat{O}_2 | \Psi_{HF} \rangle = \sum_{i<j}^{N} \left(\mathcal{J}_{ij} - \mathcal{K}_{ij} \right)$ *for the same system and write down the HF energy"*

Electron interaction in a two-electron determinant

Problem: Two-electron Hamiltonian

"Show that $\langle \Psi_{HF} | \hat{O}_2 | \Psi_{HF} \rangle = \sum_{i<j}^{N} \left(\mathcal{J}_{ij} - \mathcal{K}_{ij} \right)$ for the same system and write down the HF energy"

Solution

$$\hat{O}_2 = r_{12}^{-1}$$

$$\begin{aligned}
\langle \Psi_{\mathrm{HF}} | r_{12}^{-1} | \Psi_{\mathrm{HF}} \rangle &= \frac{1}{2} \langle \chi_1\chi_2 - \chi_2\chi_1 | r_{12}^{-1} | \chi_1\chi_2 - \chi_2\chi_1 \rangle \\
&= \frac{1}{2} \Big[\langle \chi_1\chi_2 | r_{12}^{-1} | \chi_1\chi_2 \rangle - \langle \chi_1\chi_2 | r_{12}^{-1} | \chi_2\chi_1 \rangle \\
&\quad - \langle \chi_2\chi_1 | r_{12}^{-1} | \chi_1\chi_2 \rangle + \langle \chi_2\chi_1 | r_{12}^{-1} | \chi_2\chi_1 \rangle \Big] \\
&= \frac{1}{2} \Big[\mathcal{J}_{12} - \mathcal{K}_{12} - \mathcal{K}_{12} + \mathcal{J}_{12} \Big] = \mathcal{J}_{12} - \mathcal{K}_{12}
\end{aligned}$$

Remember that $\langle \chi_2\chi_1 | r_{12}^{-1} | \chi_2\chi_1 \rangle = \langle \chi_1\chi_2 | r_{12}^{-1} | \chi_1\chi_2 \rangle$

$$\boxed{E_{\mathrm{HF}} = h_{11} + h_{22} + \mathcal{J}_{12} - \mathcal{K}_{12}}$$

Three-electron system

"Find the HF energy of a three-electron system composed of the spin orbitals χ_1, χ_2 and χ_3"

HF energy of a three-electron system

Three-electron system

"Find the HF energy of a three-electron system composed of the spin orbitals χ_1, χ_2 and χ_3"

Solution

$$\hat{O}_1 = \hat{h}(\boldsymbol{x}_1) + \hat{h}(\boldsymbol{x}_2) + \hat{h}(\boldsymbol{x}_3)$$

$$\hat{O}_2 = r_{12}^{-1} + r_{13}^{-1} + r_{23}^{-1}$$

$$\vdots$$

$$\boxed{E_{\text{HF}} = h_{11} + h_{22} + h_{33} + \mathcal{J}_{12} + \mathcal{J}_{13} + \mathcal{J}_{23} - \mathcal{K}_{12} - \mathcal{K}_{13} - \mathcal{K}_{23}}$$

Closed-shell H_2 in spatial orbitals

Two-electron example: H_2 in minimal basis

In the spin orbital basis, we have

$$E_{\mathrm{HF}} = \langle \chi_1 | \hat{h} | \chi_1 \rangle + \langle \chi_2 | \hat{h} | \chi_2 \rangle + \langle \chi_1 \chi_2 | \chi_1 \chi_2 \rangle - \langle \chi_1 \chi_2 | \chi_2 \chi_1 \rangle$$

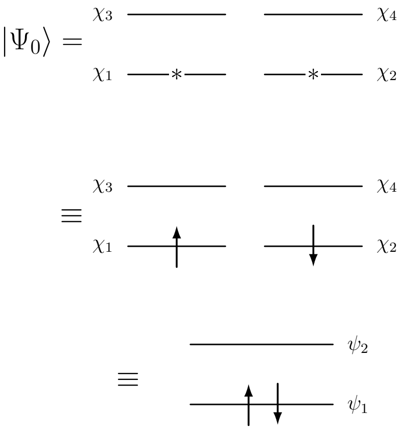
Spin to spatial transformation:

$$\chi_1(\boldsymbol{x}) \equiv \psi_1(\boldsymbol{x}) = \psi_1(\boldsymbol{r})\alpha(\omega)$$
$$\chi_2(\boldsymbol{x}) \equiv \bar{\psi}_1(\boldsymbol{x}) = \psi_1(\boldsymbol{r})\beta(\omega)$$

$$E_{\mathrm{HF}} = (\psi_1 | \hat{h} | \psi_1) + (\bar{\psi}_1 | \hat{h} | \bar{\psi}_1) + (\psi_1 \psi_1 | \bar{\psi}_1 \bar{\psi}_1) - (\psi_1 \bar{\psi}_1 | \bar{\psi}_1 \psi_1)$$

Therefore, in the spatial orbital basis, we have

$$E_{\mathrm{HF}} = 2(\psi_1 | \hat{h} | \psi_1) + (\psi_1 \psi_1 | \psi_1 \psi_1) = 2(1|\hat{h}|1) + (11|11)$$



Integrating the one-electron spin coordinate

One-electron terms

$$
\begin{aligned}
(\chi_1|\hat{h}|\chi_1) &= \int \chi_1^*(\boldsymbol{x})\hat{h}(\boldsymbol{x})\chi_1(\boldsymbol{x})\,\mathrm{d}\boldsymbol{x} \\
&= \iint \alpha^*(\omega)\psi_1^*(\boldsymbol{r})\hat{h}(\boldsymbol{r})\alpha(\omega)\psi_1(\boldsymbol{r})\,\mathrm{d}\omega\,\mathrm{d}\boldsymbol{r} \\
&= \underbrace{\left[\int \alpha^*(\omega)\alpha(\omega)\,\mathrm{d}\omega\right]}_{=1}\underbrace{\left[\int \psi_1^*(\boldsymbol{r})\hat{h}(\boldsymbol{r})\psi_1(\boldsymbol{r})\,\mathrm{d}\boldsymbol{r}\right]}_{(\psi_1|\hat{h}|\psi_1)}
\end{aligned}
$$

$$
\begin{aligned}
(\chi_2|\hat{h}|\chi_2) &= \int \chi_2^*(\boldsymbol{x})\hat{h}(\boldsymbol{x})\chi_2(\boldsymbol{x})\,\mathrm{d}\boldsymbol{x} \\
&= \iint \beta^*(\omega)\psi_1^*(\boldsymbol{r})\hat{h}(\boldsymbol{r})\beta(\omega)\psi_1(\boldsymbol{r})\,\mathrm{d}\omega\,\mathrm{d}\boldsymbol{r} \\
&= \underbrace{\left[\int \beta^*(\omega)\beta(\omega)\,\mathrm{d}\omega\right]}_{=1}\underbrace{\left[\int \psi_1^*(\boldsymbol{r})\hat{h}(\boldsymbol{r})\psi_1(\boldsymbol{r})\,\mathrm{d}\boldsymbol{r}\right]}_{(\psi_1|\hat{h}|\psi_1)}
\end{aligned}
$$

Integrating spin in Coulomb and exchange terms

Two-electron terms

$$
\begin{aligned}
(\chi_1\chi_1|\chi_2\chi_2) &= \iint \chi_1^*(\boldsymbol{x}_1)\chi_1(\boldsymbol{x}_1)r_{12}^{-1}\chi_2^*(\boldsymbol{x}_2)\chi_2(\boldsymbol{x}_2)\,\mathrm{d}\boldsymbol{x}_1\,\mathrm{d}\boldsymbol{x}_2 \\
&= \iint\iint \alpha^*(\omega_1)\psi_1^*(\boldsymbol{r}_1)\alpha(\omega_1)\psi_1(\boldsymbol{r}_1)r_{12}^{-1}\beta^*(\omega_2)\psi_1^*(\boldsymbol{r}_2)\beta(\omega_2)\psi_1(\boldsymbol{r}_2)\,\mathrm{d}\omega_1\,\mathrm{d}\boldsymbol{r}_1\,\mathrm{d}\omega_2\,\mathrm{d}\boldsymbol{r}_2 \\
&= \underbrace{\left[\int \alpha^*(\omega_1)\alpha(\omega_1)\,\mathrm{d}\omega_1\right]}_{=1}\underbrace{\left[\int \beta^*(\omega_2)\beta(\omega_2)\,\mathrm{d}\omega_2\right]}_{=1}\underbrace{\left[\iint \psi_1^*(\boldsymbol{r}_1)\psi_1(\boldsymbol{r}_1)r_{12}^{-1}\psi_1^*(\boldsymbol{r}_2)\psi_1(\boldsymbol{r}_2)\,\mathrm{d}\boldsymbol{r}_1\,\mathrm{d}\boldsymbol{r}_2\right]}_{(\psi_1\psi_1|\psi_1\psi_1)}
\end{aligned}
$$

$$
\begin{aligned}
(\chi_1\chi_2|\chi_2\chi_1) &= \iint \chi_1^*(\boldsymbol{x}_1)\chi_2(\boldsymbol{x}_1)r_{12}^{-1}\chi_2^*(\boldsymbol{x}_2)\chi_1(\boldsymbol{x}_2)\,\mathrm{d}\boldsymbol{x}_1\,\mathrm{d}\boldsymbol{x}_2 \\
&= \iint\iint \alpha^*(\omega_1)\psi_1^*(\boldsymbol{r}_1)\beta(\omega_1)\psi_1(\boldsymbol{r}_1)r_{12}^{-1}\beta^*(\omega_2)\psi_1^*(\boldsymbol{r}_2)\alpha(\omega_2)\psi_1(\boldsymbol{r}_2)\,\mathrm{d}\omega_1\,\mathrm{d}\boldsymbol{r}_1\,\mathrm{d}\omega_2\,\mathrm{d}\boldsymbol{r}_2 \\
&= \underbrace{\left[\int \alpha^*(\omega_1)\beta(\omega_1)\,\mathrm{d}\omega_1\right]}_{=0}\underbrace{\left[\int \beta^*(\omega_2)\alpha(\omega_2)\,\mathrm{d}\omega_2\right]}_{=0}\underbrace{\left[\iint \psi_1^*(\boldsymbol{r}_1)\psi_1(\boldsymbol{r}_1)r_{12}^{-1}\psi_1^*(\boldsymbol{r}_2)\psi_1(\boldsymbol{r}_2)\,\mathrm{d}\boldsymbol{r}_1\,\mathrm{d}\boldsymbol{r}_2\right]}_{(\psi_1\psi_1|\psi_1\psi_1)}
\end{aligned}
$$

Closed-shell Hartree-Fock energy in spatial orbitals

General expression

$$E_{\mathrm{HF}} = \sum_i^N \langle i|\hat{h}|i\rangle + \frac{1}{2}\sum_i^N\sum_j^N \left[\langle ij|ij\rangle - \langle ij|ji\rangle\right] = 2\sum_i^{N/2}(i|\hat{h}|i) + \sum_i^{N/2}\sum_j^{N/2}\left[2(ii|jj) - (ij|ji)\right]$$

One- and two-electron terms

$$\sum_i^N \langle i|\hat{h}|i\rangle = \sum_i^{N/2}\langle i|\hat{h}|i\rangle + \sum_i^{N/2}\langle \bar{i}|\hat{h}|\bar{i}\rangle = 2\sum_i^{N/2}(i|\hat{h}|i)$$

$$\begin{aligned}
\frac{1}{2}\sum_i^N\sum_j^N \left[\langle ij|ij\rangle - \langle ij|ji\rangle\right] = \frac{1}{2}\Bigg\{ &\sum_i^{N/2}\sum_j^{N/2}\left(\langle ij|ij\rangle - \langle ij|ji\rangle\right) + \sum_i^{N/2}\sum_j^{N/2}\left(\langle \bar{i}j|\bar{i}j\rangle - \langle \bar{i}j|j\bar{i}\rangle\right) \\
&+ \sum_i^{N/2}\sum_j^{N/2}\left(\langle i\bar{j}|i\bar{j}\rangle - \langle i\bar{j}|\bar{j}i\rangle\right) + \sum_i^{N/2}\sum_j^{N/2}\left(\langle \bar{i}\bar{j}|\bar{i}\bar{j}\rangle - \langle \bar{i}\bar{j}|\bar{j}\bar{i}\rangle\right)\Bigg\} \\
= &\sum_i^{N/2}\sum_j^{N/2}\left[2(ii|jj) - (ij|ji)\right]
\end{aligned}$$

Problem: HF energy of the Li atom

"Find the HF energy of the Li atom in terms of spatial orbitals"

Problem: HF energy of the Li atom

"Find the HF energy of the Li atom in terms of spatial orbitals"

Solution

$$\chi_1 = \alpha\psi_1 \qquad \chi_2 = \beta\psi_1 \qquad \chi_3 = \alpha\psi_2$$

$$E_{\mathrm{HF}} = 2h_{11} + h_{22} + J_{11} + 2J_{12} - K_{12}$$

Problem: HF energy of the Li atom

"Find the HF energy of the Li atom in terms of spatial orbitals"

Solution

$$\chi_1 = \alpha\psi_1 \qquad \chi_2 = \beta\psi_1 \qquad \chi_3 = \alpha\psi_2$$

$$E_{\mathrm{HF}} = 2h_{11} + h_{22} + J_{11} + 2J_{12} - K_{12}$$

Problem: HF energy of the B atom

"Find the HF energy of the B atom in terms of spatial orbitals"

Problem: HF energy of the Li atom

"Find the HF energy of the Li atom in terms of spatial orbitals"

Solution

$$\chi_1 = \alpha\psi_1 \qquad \chi_2 = \beta\psi_1 \qquad \chi_3 = \alpha\psi_2$$

$$E_{\mathrm{HF}} = 2h_{11} + h_{22} + J_{11} + 2J_{12} - K_{12}$$

Problem: HF energy of the B atom

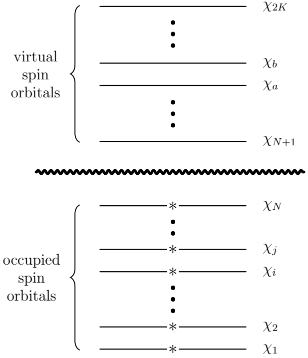
"Find the HF energy of the B atom in terms of spatial orbitals"

Solution

$$E_{\mathrm{HF}} = 2h_{11} + 2h_{22} + h_{33} + J_{11} + 4J_{12} + J_{22} - 2K_{12} + 2J_{13} + 2J_{23} - K_{13} - K_{23}$$

Reference ground-state determinant

The electrons are in the N lowest spin orbitals (Aufbau principle): $|\Psi_0\rangle = |\chi_1 \ldots \chi_i \chi_j \ldots \chi_N\rangle$



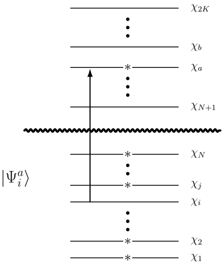
Excited determinants

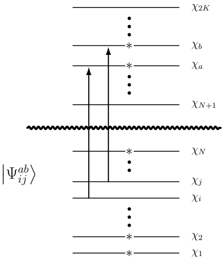
Singly-excited determinants

Electron in i promoted in a: $\quad \left|\Psi_i^a\right\rangle = \left|\chi_1 \dots \chi_a \chi_j \dots \chi_N\right\rangle = \hat{a}_a^\dagger \hat{a}_i \left|\Psi_0\right\rangle$

Doubly-excited determinants

Electrons in i and j promoted in a and b: $\quad \left|\Psi_{ij}^{ab}\right\rangle = \left|\chi_1 \dots \chi_a \chi_b \dots \chi_N\right\rangle = \hat{a}_a^\dagger \hat{a}_b^\dagger \hat{a}_j \hat{a}_i \left|\Psi_0\right\rangle$





Slater-Condon rules: One-electron operators

$$\boxed{\hat{O}_1 = \sum_i^N \hat{h}(\boldsymbol{r}_i)}$$

Case 1: determinants differ by zero spin orbitals

$$|K\rangle = |\cdots ij \cdots\rangle \quad \Rightarrow \quad \langle K|\hat{O}_1|K\rangle = \sum_i^N \langle i|\hat{h}|i\rangle$$ Example: $$\langle \Psi_0|\hat{O}_1|\Psi_0\rangle = \sum_i^N h_{ii}$$

Case 2: determinants differ by one spin orbital

$|K\rangle = |\cdots ij \cdots\rangle$ and $|L\rangle = |\cdots kj \cdots\rangle \quad \Rightarrow \quad \langle K|\hat{O}_1|L\rangle = \langle i|\hat{h}|k\rangle$ Example: $\langle \Psi_0|\hat{O}_1|\Psi_i^a\rangle = h_{ia}$

Case 3: determinants differ by two spin orbitals

$|K\rangle = |\cdots ij \ldots\rangle$ and $|L\rangle = |\cdots kl \ldots\rangle \quad \Rightarrow \quad \langle K|\hat{O}_1|L\rangle = 0$ Example: $\langle \Psi_0|\hat{O}_1|\Psi_{ij}^{ab}\rangle = 0$

Slater-Condon rules: Two-electron operators

$$\boxed{\hat{O}_2 = \sum_{i<j}^{N} r_{ij}^{-1}}$$

Case 1: determinants differ by zero spin orbitals

$$|K\rangle = |\cdots ij \cdots\rangle \quad \Rightarrow \quad \langle K|\hat{O}_2|K\rangle = \frac{1}{2}\sum_{ij}^{N} \langle ij||ij\rangle$$

Example: $\langle \Psi_0|\hat{O}_2|\Psi_0\rangle = \frac{1}{2}\sum_{ij}^{N} \langle ij||ij\rangle$

Case 2: determinants differ by one spin orbital

$|K\rangle = |\cdots ij \cdots\rangle$ and $|L\rangle = |\cdots kj \cdots\rangle \quad \Rightarrow \quad \langle K|\hat{O}_2|L\rangle = \sum_{j}^{N} \langle ij||kj\rangle$

Example: $\langle \Psi_0|\hat{O}_2|\Psi_i^a\rangle = \sum_{k}^{N} \langle ik||ak\rangle$

Case 3: determinants differ by two spin orbitals

$|K\rangle = |\cdots ij \cdots\rangle$ and $|L\rangle = |\cdots kl \cdots\rangle \quad \Rightarrow \quad \langle K|\hat{O}_2|L\rangle = \langle ij||kl\rangle$

Example: $\langle \Psi_0|\hat{O}_2|\Psi_{ij}^{ab}\rangle = \langle ij||ab\rangle$

Excited determinant vs excited HF solution

An excited determinant is

- ✓ An orbital substitution within a fixed set of reference orbitals
- ✓ A valid many-electron basis function
- ✓ A building block for CI and CC expansions

An excited determinant is generally not

- ✗ A separately optimized HF state
- ✗ A stationary point of the HF energy
- ✗ An eigenfunction of $\hat{\mathcal{S}}^2$

Brillouin's theorem

For a stationary HF reference, the Hamiltonian does not couple the reference determinant to a singly-excited determinant:

$$\boxed{\left\langle \Psi_0 \middle| \hat{\mathcal{H}}_{\text{elec}} \middle| \Psi_i^a \right\rangle = h_{ia} + \sum_k^N \langle ik||ak\rangle = f_{ia} = 0}$$

$\Longrightarrow$ Orbital excitation and state optimization are different procedures

Spin operators

In the nonrelativistic, spin-free framework, $\hat{\mathcal{H}}$ commutes with the total spin operators

$$\boxed{\left[\hat{\mathcal{H}}, \hat{S}^2\right] = 0 \qquad \text{and} \qquad \left[\hat{\mathcal{H}}, \hat{S}_z\right] = 0}$$

The eigenfunctions of $\hat{\mathcal{H}}$ may therefore be chosen as simultaneous eigenfunctions of $\hat{S}^2$ and $\hat{S}_z$:

$$\hat{S}^2 |S, M_S\rangle = S(S+1) |S, M_S\rangle \qquad \hat{S}_z |S, M_S\rangle = M_S |S, M_S\rangle$$

Spin quantum numbers

- S determines the spin multiplicity $2S+1$:
  - $S = 0$ for singlets
  - $S = 1/2$ for doublets
  - $S = 1$ for triplets
  - $\ldots$
- $M_S = -S, -S+1, \ldots, S-1, S$ is its spin projection

Are HF wavefunctions spin eigenfunctions?

For a collinear* determinant $|\Psi\rangle$ containing N^α spin-up and N^β spin-down orbitals, one has

$$\boxed{\hat{S}_z |\Psi\rangle = M_S |\Psi\rangle \quad \text{with} \quad M_S = \frac{N^\alpha - N^\beta}{2}}$$

Closed-shell RHF

Every spatial orbital is doubly occupied:

$$\hat{S}^2 |\Psi_{\text{RHF}}\rangle = 0$$

The determinant is a pure singlet

Open-shell UHF

The spin-up and spin-down spatial orbitals are different:

$$\hat{S}^2 |\Psi_{\text{UHF}}\rangle \neq S(S+1) |\Psi_{\text{UHF}}\rangle$$

in general

$\Rightarrow$ Every RHF or UHF determinant has a well-defined M_S, but only specific determinants have a well-defined S

*In a collinear determinant, the occupied spin orbitals all have spin aligned along one common quantization axis

Spin contamination in UHF

For $N^\alpha \geq N^\beta$, a UHF determinant satisfies

$$\boxed{\left\langle \hat{S}^2 \right\rangle_{\text{UHF}} = M_S(M_S+1) + N^\beta - \sum_{i=1}^{N^\alpha} \sum_{j=1}^{N^\beta} \left| S_{ij}^{\alpha\beta} \right|^2} \quad \text{with} \quad \underbrace{S_{ij}^{\alpha\beta}}_{\text{overlap}} = \left\langle \psi_i^\alpha \middle| \psi_j^\beta \right\rangle$$

Spin contamination

For a target high-spin state with $S = M_S$,

$$\Delta S^2 = \left\langle \hat{S}^2 \right\rangle_{\text{UHF}} - \left\langle \hat{S}^2 \right\rangle_{\text{exact}} \geq 0$$

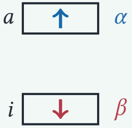
- A singlet may be contaminated by triplets, quintets, …
- A doublet may be contaminated by quartets, sextets, …
- A triplet may be contaminated by quintets, septets, …
- Restoring spin symmetry generally requires spin projection or a linear combination of determinants

Are excited determinants based on restricted orbitals spin eigenfunctions?

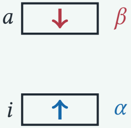
Starting from a closed-shell singlet reference $|\Psi_0\rangle$

$$|D_\alpha\rangle = \hat{a}^\dagger_{a\alpha}\hat{a}_{i\alpha}\,|\Psi_0\rangle \qquad |D_\beta\rangle = \hat{a}^\dagger_{a\beta}\hat{a}_{i\beta}\,|\Psi_0\rangle \qquad \hat{S}_z\,|D_\sigma\rangle = 0 \qquad \langle D_\sigma|\hat{S}^2|D_\sigma\rangle = 1$$

$|D_\alpha\rangle$: excite the spin-up electron



$|D_\beta\rangle$: excite the spin-down electron



Same M_S, indefinite total spin

Each determinant has $M_S = 0$, but contains equal singlet and triplet character. Being an eigenfunction of $\hat{S}_z$ does not imply being an eigenfunction of $\hat{S}^2$

$\Longrightarrow$ Open-shell determinants might not be pure spin functions!

How to get pure singlets and triplets?

In the two-determinant basis $\{ |D_\alpha\rangle, |D_\beta\rangle\}$,

$$\boxed{\left[\hat{S}^2\right]_{\{D_\alpha,D_\beta\}} = \begin{pmatrix} +1 & -1 \\ -1 & +1 \end{pmatrix}}$$

Pure singlet

$$\boxed{\left|{}^1\Psi_i^a\right\rangle = \frac{|D_\alpha\rangle + |D_\beta\rangle}{\sqrt{2}}}$$

$$\hat{S}^2 \left|{}^1\Psi_i^a\right\rangle = 0 \quad \Longrightarrow \quad S = 0$$

Pure triplet with $M_S = 0$

$$\boxed{\left|{}^{3,0}\Psi_i^a\right\rangle = \frac{|D_\alpha\rangle - |D_\beta\rangle}{\sqrt{2}}}$$

$$\hat{S}^2 \left|{}^{3,0}\Psi_i^a\right\rangle = 2 \left|{}^{3,0}\Psi_i^a\right\rangle \quad \Longrightarrow \quad S = 1$$

$\Longrightarrow$ spin adaptation = diagonalization of $\hat{S}^2$ in a determinant space

Spin and spatial symmetries

Open-shell singlet

For the two open-shell spatial orbitals i and a, the singlet is

$$|{}^1\Psi_i^a\rangle = \frac{1}{2}\underbrace{\left[\,|\phi_i(1)\phi_a(2)\rangle + |\phi_a(1)\phi_i(2)\rangle\right]}_{\text{spatially symmetric}}\underbrace{\left[\alpha(1)\beta(2) - \beta(1)\alpha(2)\right]}_{\text{spin antisymmetric}}$$

symmetric spatial part × antisymmetric spin part

Triplet with $M_S = 0$

The $M_S = 0$ triplet is

$$|{}^{3,0}\Psi_i^a\rangle = \frac{1}{2}\underbrace{\left[\,|\phi_i(1)\phi_a(2)\rangle - |\phi_a(1)\phi_i(2)\rangle\right]}_{\text{spatially antisymmetric}}\underbrace{\left[\alpha(1)\beta(2) + \beta(1)\alpha(2)\right]}_{\text{spin symmetric}}$$

antisymmetric spatial part × symmetric spin part

$\Rightarrow$ The total electronic wavefunction is antisymmetric in both cases

The three components of the triplet

Components of the $S = 1$ multiplet

$$\left|^{3,+1}\Psi_i^a\right\rangle = \left|\cdots i^\alpha a^\alpha \cdots\right\rangle \qquad \left|^{3,0}\Psi_i^a\right\rangle = \frac{|D_\alpha\rangle - |D_\beta\rangle}{\sqrt{2}} \qquad \left|^{3,-1}\Psi_i^a\right\rangle = \left|\cdots i^\beta a^\beta \cdots\right\rangle$$

Every component satisfies

$$\hat{S}^2 \left|1, M_S\right\rangle = 2 \left|1, M_S\right\rangle \qquad \hat{S}_z \left|1, M_S\right\rangle = M_S \left|1, M_S\right\rangle$$

They are connected by

$$\underbrace{\hat{S}_\pm}_{\text{ladder operator}} \left|S, M_S\right\rangle = \sqrt{S(S+1) - M_S(M_S \pm 1)} \left|S, M_S \pm 1\right\rangle$$

$M_S = \pm 1$

Each component is a single spin-pure determinant

$M_S = 0$

The pure triplet requires two determinants

The Fock operator

Hartree-Fock Lagrangian

We would like to minimize E_{HF} with the constraint that the spin orbitals remain orthonormal:

$$\mathcal{L}[\{\chi_i\}] = E_{\text{HF}}[\{\chi_i\}] - \sum_{ij} \underbrace{\epsilon_{ji}}_{\substack{\text{Lagrange} \\ \text{multipliers}}} \underbrace{\left(\langle \chi_i | \chi_j \rangle - \delta_{ij}\right)}_{\substack{\text{orthonormality} \\ \text{constraint}}}$$

The stationarity of the HF Lagrangian with respect to the spin orbitals implies that

$$\frac{\delta \mathcal{L}}{\delta \chi_i^*(\boldsymbol{x})} = 0 \quad \Rightarrow \quad \boxed{\hat{f}(\boldsymbol{x}) \chi_i(\boldsymbol{x}) = \sum_j \epsilon_{ji} \chi_j(\boldsymbol{x})}$$

One-electron Fock operator

$$\boxed{\hat{f}(\boldsymbol{x}) = \hat{h}(\boldsymbol{x}) + \underbrace{\sum_i^N [\hat{\mathcal{J}}_i(\boldsymbol{x}) - \hat{\mathcal{K}}_i(\boldsymbol{x})]}_{\hat{v}^{\text{HF}}(\boldsymbol{x}) \, = \, \text{Hartree-Fock potential}}}$$

Canonical Hartree-Fock equations

There exists a unitary transformation $\boldsymbol{U}$ such that $\widetilde{\boldsymbol{\epsilon}} = \boldsymbol{U}^{\dagger} \cdot \boldsymbol{\epsilon} \cdot \boldsymbol{U}$ is diagonal:

$$\widetilde{\chi}_i(\boldsymbol{x}) = \sum_j \chi_j(\boldsymbol{x}) U_{ji} \quad \Longrightarrow \quad |\widetilde{\Psi}_{\text{HF}}\rangle = \det(\boldsymbol{U}) \, |\Psi_{\text{HF}}\rangle = e^{i\theta} \, |\Psi_{\text{HF}}\rangle \quad \text{and} \quad \boxed{\widetilde{f}(\boldsymbol{x}) = \hat{f}(\boldsymbol{x})}$$

$$\boldsymbol{U}^{\dagger} \cdot \boldsymbol{U} = \mathbf{1} \quad \Longrightarrow \quad \det\left(\boldsymbol{U}^{\dagger} \cdot \boldsymbol{U}\right) = 1 \quad \Longrightarrow \quad |\det(\boldsymbol{U})|^2 = 1 \quad \Longrightarrow \quad \det(\boldsymbol{U}) = e^{i\theta}$$

Canonical HF equations

Dropping the tildes from hereon, we get

$$\boxed{\hat{f}(\boldsymbol{x}) \chi_i(\boldsymbol{x}) = \epsilon_i \chi_i(\boldsymbol{x})}$$

- Eigenfunctions $\{\chi_i\}$ are called canonical spin orbitals
- Eigenvalues $\{\epsilon_i\}$ are called orbital energies

Fock matrix elements in the spin orbital basis

Problem:

“ Find the expression of the matrix elements $f_{ij} = \langle \chi_i | \hat{f} | \chi_j \rangle$”

Fock matrix elements in the spin orbital basis

Problem:

" Find the expression of the matrix elements $f_{ij} = \langle \chi_i | \hat{f} | \chi_j \rangle$"

Solution

$$
\begin{aligned}
\langle \chi_i | \hat{f} | \chi_j \rangle &= \langle \chi_i | \hat{h} + \sum_k \left(\hat{\mathcal{J}}_k - \hat{\mathcal{K}}_k \right) | \chi_j \rangle \\
&= \langle \chi_i | \hat{h} | \chi_j \rangle + \sum_k \left(\langle \chi_i | \hat{\mathcal{J}}_k | \chi_j \rangle - \langle \chi_i | \hat{\mathcal{K}}_k | \chi_j \rangle \right) \\
&= \langle i | \hat{h} | j \rangle + \sum_k \left[\langle ik | jk \rangle - \langle ik | kj \rangle \right] \\
&= \langle i | \hat{h} | j \rangle + \sum_k \langle ik || jk \rangle
\end{aligned}
$$

Orbital energies in the spin orbital basis

Problem:

" Deduce the expression of ϵ_i"

Orbital energies in the spin orbital basis

Problem:

" Deduce the expression of ϵ_i"

Solution

$$
\begin{aligned}
\hat{f}\,|\chi_i\rangle = \epsilon_i\,|\chi_i\rangle \Rightarrow \quad & \langle\chi_i|\hat{f}|\chi_i\rangle = \epsilon_i\,\langle\chi_i|\chi_i\rangle = \epsilon_i \\
\Rightarrow \quad & \epsilon_i = \langle i|\hat{h}|i\rangle + \sum_k \left[\langle ik|ik\rangle - \langle ik|ki\rangle\right] \\
\Rightarrow \quad & \epsilon_i = \langle i|\hat{h}|i\rangle + \sum_k \langle ik||ik\rangle
\end{aligned}
$$

Koopmans' theorem

Ground-state energy of the N-electron system

$$ {}^{N}E_0 = \sum_i h_{ii} + \frac{1}{2} \sum_{ij} \langle ij||ij \rangle $$

Energy of the $(N-1)$-electron system (cation)

$$ {}^{N-1}E_k = \sum_{i \neq k} h_{ii} + \frac{1}{2} \sum_{i \neq k} \sum_{j \neq k} \langle ij||ij \rangle $$

Ionization potential (IP)

$$ \begin{aligned} \mathrm{IP} &\approx {}^{N-1}E_k - {}^{N}E_0 \\ &= -\langle k|\hat{h}|k \rangle - \frac{1}{2} \sum_i \langle ik||ik \rangle - \frac{1}{2} \sum_j \langle kj||kj \rangle \\ &= -\langle k|\hat{h}|k \rangle - \sum_i \langle ik||ik \rangle = -\epsilon_k \end{aligned} $$

Koopmans' theorem for electron affinity (EA)

Problem:

"Show that Koopmans' theorem applies to electron affinities"

Koopmans' theorem for electron affinity (EA)

Problem:

"Show that Koopmans' theorem applies to electron affinities"

Solution

$$\begin{aligned}\text{EA} &\approx {}^{N}E_0 - {}^{N+1}E^c \\ &= -\langle c|\hat{h}|c\rangle - \sum_i \langle ci||ci\rangle \\ &= -\epsilon_c\end{aligned}$$

Koopmans prefers IPs

Ionization potentials

$$\begin{aligned}\mathrm{IP} &= E^{N-1}_{\mathrm{exact}} - E^{N}_{\mathrm{exact}} \\ &= E^{N-1}_{\mathrm{HF}} - E^{N}_{\mathrm{HF}} + E^{N-1}_{\mathrm{corr}} - E^{N}_{\mathrm{corr}} \\ &= -\epsilon_i + \underbrace{\Delta^{N-1}_{\mathrm{relax}}}_{<0} + \underbrace{\Delta_{\mathrm{corr}}}_{>0}\end{aligned}$$

✓ Relaxation and correlation
partially cancel

Electron affinities

$$\begin{aligned}\mathrm{EA} &= E^{N}_{\mathrm{exact}} - E^{N+1}_{\mathrm{exact}} \\ &= E^{N}_{\mathrm{HF}} - E^{N+1}_{\mathrm{HF}} + E^{N}_{\mathrm{corr}} - E^{N+1}_{\mathrm{corr}} \\ &= -\epsilon_a + \underbrace{\Delta^{N+1}_{\mathrm{relax}}}_{>0} + \underbrace{\Delta_{\mathrm{corr}}}_{>0}\end{aligned}$$

✗ Relaxation and correlation
do not cancel

$\Rightarrow$ Koopmans' theorem is usually more accurate for IPs than for EAs

Hartree-Fock Hamiltonian

Definition

$$\underbrace{\hat{\mathcal{H}}^{\text{HF}}}_{N\text{-electron operator}} = \sum_i^N \underbrace{\hat{f}(\boldsymbol{x}_i)}_{\text{one-electron operator}} = \sum_i^N \left[\hat{h}(\boldsymbol{x}_i) + \hat{v}^{\text{HF}}(\boldsymbol{x}_i)\right]$$

Warnings!

$$\hat{\mathcal{H}}^{\text{HF}} |\Psi_{\text{HF}}\rangle = \left(\sum_i^N \epsilon_i\right) |\Psi_{\text{HF}}\rangle \neq E_{\text{HF}} |\Psi_{\text{HF}}\rangle$$

The HF wavefunction is an eigenfunction of the HF Hamiltonian, but the eigenvalue is not the HF energy!

$$\hat{\mathcal{H}} |\Psi_{\text{HF}}\rangle \neq E_{\text{HF}} |\Psi_{\text{HF}}\rangle$$

The HF wavefunction is not an eigenfunction of the electronic Hamiltonian!

$$\langle \Psi_{\text{HF}} | \hat{\mathcal{H}}_{\text{elec}} | \Psi_{\text{HF}} \rangle = E_{\text{HF}}$$

The HF energy is the expectation value of the HF wavefunction of the electronic Hamiltonian

Double counting

$$\sum_{i}^{N} \epsilon_i = \sum_{i}^{N} h_{ii} + \sum_{ij}^{N} (\mathcal{J}_{ij} - \mathcal{K}_{ij})$$

$$\nparallel$$

$$E_{\mathrm{HF}} = \sum_{i}^{N} h_{ii} + \frac{1}{2} \sum_{ij}^{N} (\mathcal{J}_{ij} - \mathcal{K}_{ij})$$

One double-counts the electron-electron interaction!

$$\langle \Psi_{\mathrm{HF}} | \sum_{i<j} \frac{1}{r_{ij}} - \sum_{i}^{N} v^{\mathrm{HF}}(\boldsymbol{x}_i) | \Psi_{\mathrm{HF}} \rangle = -\frac{1}{2} \sum_{ij}^{N} (\mathcal{J}_{ij} - \mathcal{K}_{ij})$$

This is the first-order correction in Møller-Plesset perturbation theory!

Roothaan-Hall equations: introduction of a basis

Expansion in a basis

$$\psi_i(\boldsymbol{r}) = \sum_{\mu}^{K} C_{\mu i}\phi_\mu(\boldsymbol{r}) \qquad \equiv \qquad |i\rangle = \sum_{\mu}^{K} C_{\mu i} |\mu\rangle$$

Roothaan-Hall equations

$$\hat{f}|i\rangle = \epsilon_i |i\rangle \quad \Rightarrow \quad \hat{f}\sum_{\nu} C_{\nu i}|\nu\rangle = \epsilon_i \sum_{\nu} C_{\nu i}|\nu\rangle$$

$$\xRightarrow{\langle\mu|} \quad \sum_{\nu} C_{\nu i} \underbrace{\langle\mu|\hat{f}|\nu\rangle}_{F_{\mu\nu}} = \epsilon_i \sum_{\nu} C_{\nu i} \underbrace{\langle\mu|\nu\rangle}_{S_{\mu\nu}}$$

$$\Rightarrow \quad \boxed{\sum_{\nu} F_{\mu\nu}C_{\nu i} = \sum_{\nu} S_{\mu\nu}C_{\nu i}\epsilon_i} \quad \Rightarrow \quad \boldsymbol{F}\cdot\boldsymbol{C} = \boldsymbol{S}\cdot\boldsymbol{C}\cdot\boldsymbol{E} \quad \text{in matrix form}$$

The Roothaan-Hall matrix equations

Matrix form of the Roothaan-Hall equations

$$\boldsymbol{F}\cdot\boldsymbol{C}=\boldsymbol{S}\cdot\boldsymbol{C}\cdot\boldsymbol{E} \qquad \Leftrightarrow \qquad \boldsymbol{F}'\cdot\boldsymbol{C}'=\boldsymbol{C}'\cdot\boldsymbol{E}$$

$$\boldsymbol{F}'=\boldsymbol{X}^{\dagger}\cdot\boldsymbol{F}\cdot\boldsymbol{X} \qquad \boldsymbol{C}=\boldsymbol{X}\cdot\boldsymbol{C}' \qquad \boldsymbol{X}^{\dagger}\cdot\boldsymbol{S}\cdot\boldsymbol{X}=\boldsymbol{1}$$

- Fock matrix $F_{\mu\nu}=\langle\mu|\hat{f}|\nu\rangle$ and overlap matrix $S_{\mu\nu}=\langle\mu|\nu\rangle$
- We need to determine the coefficient matrix $\boldsymbol{C}$ and the orbital energies $\boldsymbol{E}$

$$\boldsymbol{C}=\begin{pmatrix} C_{11} & C_{12} & \cdots & C_{1K} \\ C_{21} & C_{22} & \cdots & C_{2K} \\ \vdots & \vdots & \ddots & \vdots \\ C_{K1} & C_{K2} & \cdots & C_{KK} \end{pmatrix} \qquad \boldsymbol{E}=\begin{pmatrix} \epsilon_1 & 0 & \cdots & 0 \\ 0 & \epsilon_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \epsilon_K \end{pmatrix}$$

Self-consistent field (SCF) procedure

$$\boldsymbol{F}(\boldsymbol{C})\cdot\boldsymbol{C}=\boldsymbol{S}\cdot\boldsymbol{C}\cdot\boldsymbol{E}$$

How do we solve these HF equations?

From generalized to standard

$$\boldsymbol{F} \cdot \boldsymbol{C} = \underbrace{\boldsymbol{S}}_{\text{metric}} \cdot \boldsymbol{C} \cdot \boldsymbol{E} \quad \Leftrightarrow \quad \text{Generalized eigenvalue problem}$$

$$\Downarrow$$

$$\boldsymbol{X}^{\dagger} \cdot \boldsymbol{F} \cdot \underbrace{\boldsymbol{C}}_{\boldsymbol{X} \cdot \boldsymbol{C}'} = \boldsymbol{X}^{\dagger} \cdot \boldsymbol{S} \cdot \underbrace{\boldsymbol{C}}_{\boldsymbol{X} \cdot \boldsymbol{C}'} \cdot \boldsymbol{E}$$

$$\Downarrow$$

$$\underbrace{\boldsymbol{X}^{\dagger} \cdot \boldsymbol{F} \cdot \boldsymbol{X}}_{\boldsymbol{F}'} \cdot \boldsymbol{C}' = \underbrace{\boldsymbol{X}^{\dagger} \cdot \boldsymbol{S} \cdot \boldsymbol{X}}_{\boldsymbol{1}} \cdot \boldsymbol{C}' \cdot \boldsymbol{E}$$

$$\Downarrow$$

$$\boldsymbol{F}' \cdot \boldsymbol{C}' = \boldsymbol{C}' \cdot \boldsymbol{E} \quad \Leftrightarrow \quad \text{Standard eigenvalue problem}$$

Expression of the Fock matrix

Problem:

“Find the expression of the Fock matrix in terms of the one- and two-electron integrals”

Expression of the Fock matrix

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"Find the expression of the Fock matrix in terms of the one- and two-electron integrals"

Solution

$$F_{\mu\nu} = \langle \mu | \hat{h} + \sum_i^N (\hat{\mathcal{J}}_i - \hat{\mathcal{K}}_i) | \nu \rangle = \underbrace{\langle \mu | \hat{h} | \nu \rangle}_{H_{\mu\nu}} + \sum_i^N \langle \mu | \hat{\mathcal{J}}_i - \hat{\mathcal{K}}_i | \nu \rangle$$

$$= H_{\mu\nu} + \sum_i^N \Big(\langle \mu \chi_i | r_{12}^{-1} | \nu \chi_i \rangle - \langle \mu \chi_i | r_{12}^{-1} | \chi_i \nu \rangle \Big)$$

$$= H_{\mu\nu} + \sum_i^N \sum_{\lambda\sigma} C_{\lambda i} C_{\sigma i}^* \Big(\langle \mu\sigma | r_{12}^{-1} | \nu\lambda \rangle - \langle \mu\sigma | r_{12}^{-1} | \lambda\nu \rangle \Big)$$

$$= H_{\mu\nu} + \sum_{\lambda\sigma} P_{\lambda\sigma} (\langle \mu\sigma | \nu\lambda \rangle - \langle \mu\sigma | \lambda\nu \rangle)$$

$$= \underbrace{H_{\mu\nu} + \sum_{\lambda\sigma} P_{\lambda\sigma} \langle \mu\sigma || \nu\lambda \rangle}_{G_{\mu\nu}} = H_{\mu\nu} + G_{\mu\nu}$$

Core Hamiltonian

$$H_{\mu\nu} = \langle \mu | \hat{h} | \nu \rangle$$

Density matrix

$$P_{\lambda\sigma} = \sum_i^N C_{\lambda i} C_{\sigma i}^*$$

Two-electron part

$$G_{\mu\nu} = \sum_{\lambda\sigma} P_{\lambda\sigma} \langle \mu\sigma || \nu\lambda \rangle$$

Expression of the HF energy

Problem:

"Find the expression of the HF energy in terms of the one- and two-electron integrals"

Expression of the HF energy

Problem:

"Find the expression of the HF energy in terms of the one- and two-electron integrals"

Solution

$$
\begin{aligned}
E_{\mathrm{HF}} &= \sum_{i}^{N} h_{ii} + \frac{1}{2} \sum_{ij}^{N} \left(\mathcal{J}_{ij} - \mathcal{K}_{ij} \right) \\
&= \sum_{i}^{N} \left\langle \sum_{\mu} C_{\mu i}^{*} \phi_{\mu} \middle| \hat{h} \middle| \sum_{\nu} C_{\nu i} \phi_{\nu} \right\rangle + \frac{1}{2} \sum_{ij}^{N} \left\langle \left(\sum_{\mu} C_{\mu i}^{*} \phi_{\mu} \right) \left(\sum_{\lambda} C_{\lambda j}^{*} \phi_{\lambda} \right) \middle\| \left(\sum_{\nu} C_{\nu i} \phi_{\nu} \right) \left(\sum_{\sigma} C_{\sigma j} \phi_{\sigma} \right) \right\rangle \\
&= \sum_{\mu\nu} P_{\nu\mu} \left(H_{\mu\nu} + \frac{1}{2} \sum_{\lambda\sigma} P_{\sigma\lambda} \langle \mu\lambda || \nu\sigma \rangle \right) \\
&= \frac{1}{2} \sum_{\mu\nu} P_{\nu\mu} \left(H_{\mu\nu} + F_{\mu\nu} \right) = \frac{1}{2} \operatorname{Tr} \left[\boldsymbol{P} \cdot (\boldsymbol{H} + \boldsymbol{F}) \right]
\end{aligned}
$$

Computation of the Fock matrix and energy

Density matrix (closed-shell system)

$$P_{\mu\nu} = 2 \sum_{i=1}^{N/2} C_{\mu i} C_{\nu i}^{*} \quad \text{or} \quad \boxed{\boldsymbol{P} = 2\boldsymbol{C}_{\text{o}} \cdot \boldsymbol{C}_{\text{o}}^{\dagger}}$$

Fock matrix in the AO basis (closed-shell system)

$$F_{\mu\nu} = H_{\mu\nu} + \underbrace{\sum_{\lambda\sigma} P_{\sigma\lambda} (\mu\nu|\lambda\sigma)}_{J_{\mu\nu} = \text{Coulomb}} \underbrace{- \frac{1}{2} \sum_{\lambda\sigma} P_{\sigma\lambda} (\mu\sigma|\lambda\nu)}_{K_{\mu\nu} = \text{exchange}}$$

HF energy in the AO basis (closed-shell system)

$$E_{\text{HF}} = \sum_{\mu\nu} P_{\nu\mu} H_{\mu\nu} + \frac{1}{2} \sum_{\mu\nu\lambda\sigma} P_{\nu\mu} \left[(\mu\nu|\lambda\sigma) - \frac{1}{2} (\mu\sigma|\lambda\nu) \right] P_{\sigma\lambda}$$

How to perform a RHF calculation in practice

The SCF algorithm

1. Specify molecule $\{\boldsymbol{R}_A\}$ and $\{Z_A\}$ and basis set $\{\phi_\mu\}$
2. Calculate integrals $S_{\mu\nu}$, $H_{\mu\nu}$ and $\langle \mu\nu | \lambda\sigma \rangle$
3. Diagonalize $\boldsymbol{S}$ and compute $\boldsymbol{X}$
4. Choose an initial density matrix $\boldsymbol{P}$
    1. Calculate $\boldsymbol{G}$ and then $\boldsymbol{F} = \boldsymbol{H} + \boldsymbol{G}$
    2. Compute $\boldsymbol{F}' = \boldsymbol{X}^\dagger \cdot \boldsymbol{F} \cdot \boldsymbol{X}$
    3. Diagonalize $\boldsymbol{F}'$ to obtain $\boldsymbol{C}'$ and $\boldsymbol{E}$
    4. Calculate $\boldsymbol{C} = \boldsymbol{X} \cdot \boldsymbol{C}'$
    5. Form a new density matrix $\boldsymbol{P} = 2\boldsymbol{C}_\text{o} \cdot \boldsymbol{C}_\text{o}^\dagger$
    6. Am I converged? If not go back to 1.
5. Evaluate the desired observables, including E_HF

Orthogonalization matrix

We seek a matrix that orthogonalizes the AO basis, i.e., $\boldsymbol{X}^\dagger \cdot \boldsymbol{S} \cdot \boldsymbol{X} = \mathbf{1}$

Symmetric (or Löwdin) orthogonalization

$$\boldsymbol{X} = \boldsymbol{S}^{-1/2} = \boldsymbol{U} \cdot \boldsymbol{s}^{-1/2} \cdot \boldsymbol{U}^\dagger \text{ is one solution}$$

Is it working?

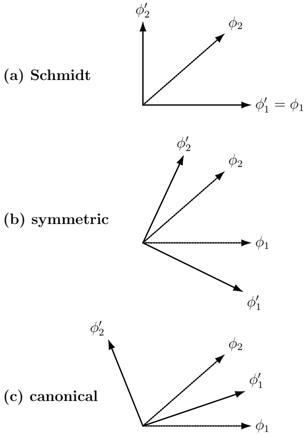
$$\boldsymbol{X}^\dagger \cdot \boldsymbol{S} \cdot \boldsymbol{X} = \boldsymbol{S}^{-1/2} \cdot \boldsymbol{S} \cdot \boldsymbol{S}^{-1/2} = \mathbf{1} \quad \checkmark$$

Canonical orthogonalization

$\boldsymbol{X} = \boldsymbol{U} \cdot \boldsymbol{s}^{-1/2}$ is another solution, useful when linear dependencies are present

Is it working?

$$\boldsymbol{X}^\dagger \cdot \boldsymbol{S} \cdot \boldsymbol{X} = \boldsymbol{s}^{-1/2} \cdot \underbrace{\boldsymbol{U}^\dagger \cdot \boldsymbol{S} \cdot \boldsymbol{U}}_{\boldsymbol{s}} \cdot \boldsymbol{s}^{-1/2} = \mathbf{1} \quad \checkmark$$



How to obtain a good guess for the MOs or density matrix?

Possible initial density matrix

1. We can set $\boldsymbol{P} = \boldsymbol{0} \Rightarrow \boldsymbol{F} = \boldsymbol{H}$ (core guess):
   $\Rightarrow$ Usually a poor guess but easy to implement
2. Use extended Hückel theory or a semi-empirical method:
   $\Rightarrow$ Inexpensive, though now less common
3. Use tabulated atomic densities:
   $\Rightarrow$ Superposition-of-atomic-densities (SAD) guess
4. Read the MOs of a previous calculation:
   $\Rightarrow$ Very common and very useful

How do I know I have converged (or not)?

The SCF procedure defines an iterative map

$$\boxed{\boldsymbol{P}^{(n)} \longrightarrow \boldsymbol{F}[\boldsymbol{P}^{(n)}] \longrightarrow \boldsymbol{P}^{(n+1)}}$$

At convergence,

$$\boldsymbol{P}^{(n+1)} = \boldsymbol{P}^{(n)} = \boldsymbol{P}^{\star}$$

1. One can check the energy and/or the density matrix
2. At self-consistency, the generalized commutator vanishes:

$$\boxed{\boldsymbol{e}^{(n)} = \boldsymbol{F}^{(n)} \cdot \boldsymbol{P}^{(n)} \cdot \boldsymbol{S} - \boldsymbol{S} \cdot \boldsymbol{P}^{(n)} \cdot \boldsymbol{F}^{(n)}}$$

$$\boldsymbol{P}^{(n)} = \boldsymbol{P}^{\star} \qquad \Longrightarrow \qquad \boldsymbol{e}^{(n)} = \boldsymbol{0} \qquad \Longrightarrow \qquad \max|\boldsymbol{e}^{(n)}| < \tau_{\text{SCF}}$$

3. DIIS (direct inversion in the iterative subspace) is usually used to speed up convergence:

$\Rightarrow$ Extrapolation of the Fock matrix using the m previous iterations: $$\boxed{\boldsymbol{F}^{(n+1)} = \sum_{i=n-m}^{n} c_i \boldsymbol{F}^{(i)}}$$

How does DIIS work?

Using the m stored iterations, DIIS constructs

$$\boldsymbol{F}_{\text{DIIS}}^{(m+1)} = \sum_{i=1}^{m} c_i \boldsymbol{F}^{(i)} \qquad \boldsymbol{e}_{\text{DIIS}}^{(m+1)} \approx \sum_{i=1}^{m} c_i \boldsymbol{e}^{(i)}$$

The coefficients minimize the residual norm:

$$\boxed{\min_{\boldsymbol{c}} \left\| \sum_{i=1}^{m} c_i \boldsymbol{e}^{(i)} \right\|_F^2 \qquad \text{subject to} \qquad \sum_{i=1}^{m} c_i = 1}$$

Define the DIIS overlap matrix

$$B_{ij} = \langle \boldsymbol{e}^{(i)} | \boldsymbol{e}^{(j)} \rangle = \operatorname{Tr}\left[\boldsymbol{e}^{(i)\dagger} \cdot \boldsymbol{e}^{(j)} \right]$$

Then

$$\left\| \boldsymbol{e}_{\text{DIIS}}^{(m+1)} \right\|_F^2 = \sum_{ij} c_i B_{ij} c_j = \boldsymbol{c}^{\mathsf{T}} \cdot \boldsymbol{B} \cdot \boldsymbol{c}$$

DIIS weights and extrapolated Fock matrix

DIIS Lagrangian

$$\mathcal{L}(\boldsymbol{c}, \lambda) = \boldsymbol{c}^{\mathsf{T}} \cdot \boldsymbol{B} \cdot \boldsymbol{c} - 2\lambda \left(\sum_{i=1}^{m} c_i - 1 \right)$$

Stationarity conditions

$$\frac{\partial \mathcal{L}(\boldsymbol{c}, \lambda)}{\partial \boldsymbol{c}} = \boldsymbol{0} \quad \text{and} \quad \frac{\partial \mathcal{L}(\boldsymbol{c}, \lambda)}{\partial \lambda} = 0 \quad \Rightarrow \quad \boldsymbol{B} \cdot \boldsymbol{c} - \lambda \boldsymbol{1} = \boldsymbol{0} \quad \text{and} \quad \sum_{i=1}^{m} c_i = 1$$

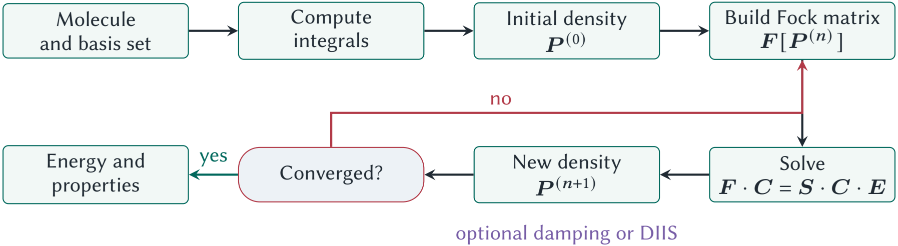
DIIS linear system

$$\begin{pmatrix} B_{11} & \cdots & B_{1m} & -1 \\ \vdots & \ddots & \vdots & \vdots \\ B_{m1} & \cdots & B_{mm} & -1 \\ -1 & \cdots & -1 & 0 \end{pmatrix} \cdot \begin{pmatrix} c_1 \\ \vdots \\ c_m \\ \lambda \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ -1 \end{pmatrix} \quad \Rightarrow \quad \boxed{\boldsymbol{F}_{\text{DIIS}}^{(m+1)} = \sum_{i=1}^{m} c_i \boldsymbol{F}^{(i)}}$$

$\Rightarrow$ The linear system has dimension only $(m+1) \times (m+1)$

DIIS algorithm

1. Start from the current density $\boldsymbol{P}^{(n)}$
2. Build the corresponding Fock matrix $\boldsymbol{F}^{(n)} = \boldsymbol{F}[\boldsymbol{P}^{(n)}]$
3. Evaluate the commutator residual $\boldsymbol{e}^{(n)} = \boldsymbol{F}^{(n)} \cdot \boldsymbol{P}^{(n)} \cdot \boldsymbol{S} - \boldsymbol{S} \cdot \boldsymbol{P}^{(n)} \cdot \boldsymbol{F}^{(n)}$
4. Store the pair $\{\boldsymbol{F}^{(n)}, \boldsymbol{e}^{(n)}\}$ and update the DIIS subspace
5. Construct $\boldsymbol{B}$, solve the augmented linear system, and obtain the coefficients $\{c_i\}$
6. Extrapolate the Fock matrix $\boldsymbol{F}^{\mathrm{DIIS}} = \sum_i c_i \boldsymbol{F}^{(i)}$
7. Solve $\boldsymbol{F}^{\mathrm{DIIS}} \cdot \boldsymbol{C} = \boldsymbol{S} \cdot \boldsymbol{C} \cdot \boldsymbol{E}$ and form the next density $\boldsymbol{P}^{(n+1)}$

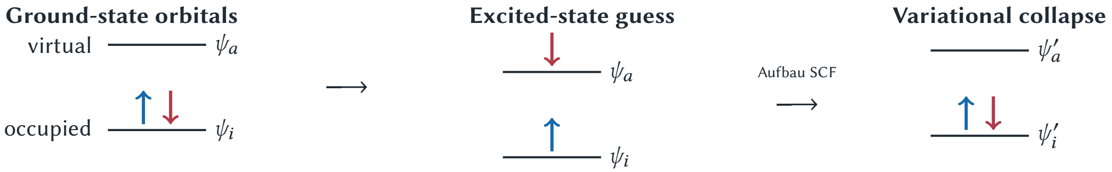


DIIS take-home messages

Key idea: DIIS accelerates SCF convergence by finding a linear combination of Fock matrices

- ✓ A small rolling subspace is normally sufficient: commonly $m \approx 6$–10
- ✗ DIIS requires additional storage: $\sim 2mK^2$
- ✗ The extrapolated energy is not guaranteed to decrease monotonically
- ✗ $\boldsymbol{B}$ can be singular due to nearly linear dependence of the residuals: a restart may be required
- ✓ DIIS can be used for many other methods!

An excited determinant can be generated by promoting one or more electrons, but the usual Aufbau occupation does not preserve that configuration.



The problem

After each diagonalization, Aufbau occupies the lowest-energy orbitals
$\Rightarrow$ the SCF tends to return to a lower-energy solution

Maximum overlap method (MOM)

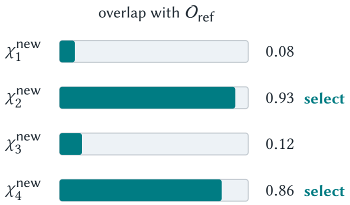
Let $C_{\mathrm{o}}^{\mathrm{ref}}$ contain the coefficients of the occupied reference spin orbitals defining the target configuration.

$$\forall \chi_p^{\mathrm{new}} \quad \text{compute} \quad O_{ip} = \left\langle \chi_i^{\mathrm{ref}} \middle| \chi_p^{\mathrm{new}} \right\rangle = \sum_{\mu\nu} C_{\mu i}^{\mathrm{ref}*} S_{\mu\nu} C_{\nu p}^{\mathrm{new}} \quad \text{and} \quad \boxed{w_p = \sum_{i \in O_{\mathrm{ref}}} \left| O_{ip} \right|^2}$$

MOM procedure

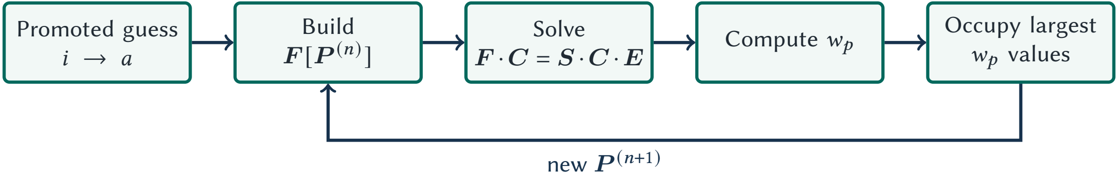
- Compute w_p for every new orbital
- Occupy the N spin orbitals with the largest w_p
- In MOM, the reference occupied space comes from the previous iteration
- In IMOM, the reference orbitals remain fixed to the initial guess

Occupied reference space
$O_{\mathrm{ref}} = \mathrm{span}\{\chi_i^{\mathrm{ref}}\}$



Select by orbital character, not by energy!

MOM-based SCF



Conventional SCF

Occupy the lowest orbital energies

$$\epsilon_1 \leq \epsilon_2 \leq \cdots \leq \epsilon_N$$

$\Longrightarrow$ finds a low-energy solution

MOM-SCF

Occupy the largest w values

$$w_1 \geq w_2 \geq \cdots \geq w_N$$

$\Longrightarrow$ follows a chosen SCF solution

MOM changes the occupation to target non-Aufbau stationary solutions of the HF equations

What does a MOM calculation give?

What is gained?

- ✓ A higher-energy stationary solution of the HF equations
- ✓ Useful excitation energies for states dominated by a single electronic configuration
- ✓ Full orbital relaxation for the targeted configuration
- ✓ A state-specific density and molecular properties
- ✓ The same formal cost per SCF iteration, although convergence may be more difficult

What are the cautions?

- ✗ The result is still a single determinant
- ✗ Spin symmetry may be broken
- ✗ Different SCF states are not necessarily orthogonal
- ✗ Convergence depends on the initial promoted orbitals
- ✗ MOM can drift or converge to an unintended solution

Dipole moments

Classical vs Quantum

$$\boldsymbol{\mu} = (\mu_x, \mu_y, \mu_z) = \underbrace{\sum_i q_i \boldsymbol{r}_i}_{\text{classical definition}}$$

$$\boldsymbol{\mu} = (\mu_x, \mu_y, \mu_z) = \underbrace{\langle \Psi_0 | -\sum_i^N \boldsymbol{r}_i | \Psi_0 \rangle}_{\text{electrons}} + \underbrace{\sum_A^M Z_A \boldsymbol{R}_A}_{\text{nuclei}} = -\sum_{\mu\nu} P_{\mu\nu} (\nu | \boldsymbol{r} | \mu) + \sum_A^M Z_A \boldsymbol{R}_A$$

Vector components

$$\mu_x = -\sum_{\mu\nu} P_{\mu\nu} (\nu | x | \mu) + \sum_A^M Z_A X_A \quad \text{with} \quad \underbrace{(\nu | x | \mu)}_{\text{one-electron integrals}} = \int \phi_\nu^*(\boldsymbol{r})\, x\, \phi_\mu(\boldsymbol{r})\, \mathrm{d}\boldsymbol{r}$$

Charge analysis

Electron density

$$\rho(\boldsymbol{r}) = \sum_{\mu\nu} \phi_\mu(\boldsymbol{r}) P_{\mu\nu} \phi_\nu^*(\boldsymbol{r}) \quad \text{with} \quad \int \rho(\boldsymbol{r})\,\mathrm{d}\boldsymbol{r} = N \quad \Rightarrow \quad N = \sum_{\mu\nu} P_{\mu\nu} S_{\nu\mu} = \sum_{\mu} (\boldsymbol{P}\cdot\boldsymbol{S})_{\mu\mu} = \mathrm{Tr}(\boldsymbol{P}\cdot\boldsymbol{S})$$

Mulliken population analysis

Assuming that the basis functions are atom-centered

$$\underbrace{q_A^{\text{Mulliken}}}_{\text{net charge on } A} = Z_A - \sum_{\mu\in A} (\boldsymbol{P}\cdot\boldsymbol{S})_{\mu\mu}$$

Löwdin population analysis

Because $\mathrm{Tr}(\boldsymbol{A}\cdot\boldsymbol{B}) = \mathrm{Tr}(\boldsymbol{B}\cdot\boldsymbol{A})$, we have, for any α, $N = \sum_\mu (\boldsymbol{S}^\alpha \cdot \boldsymbol{P} \cdot \boldsymbol{S}^{1-\alpha})_{\mu\mu}$

For $\alpha = 1/2$, we get: $\quad N = \sum_{\mu} (\boldsymbol{S}^{1/2}\cdot\boldsymbol{P}\cdot\boldsymbol{S}^{1/2})_{\mu\mu} \quad \Rightarrow \quad q_A^{\text{Löwdin}} = Z_A - \sum_{\mu\in A} (\boldsymbol{S}^{1/2}\cdot\boldsymbol{P}\cdot\boldsymbol{S}^{1/2})_{\mu\mu}$

Localized orbitals

Localized orbitals

$$f = \begin{pmatrix} f_{\mathrm{oo}} & f_{\mathrm{ov}} \\ f_{\mathrm{vo}} & f_{\mathrm{vv}} \end{pmatrix} \xrightarrow{\text{HF convergence}} \begin{pmatrix} f_{\mathrm{oo}} & 0 \\ 0 & f_{\mathrm{vv}} \end{pmatrix} \xrightarrow{\text{canonicalization}} \begin{pmatrix} \epsilon_{\mathrm{o}} & 0 \\ 0 & \epsilon_{\mathrm{v}} \end{pmatrix} \xrightarrow{\text{localization}} \begin{pmatrix} \widetilde{f}_{\mathrm{oo}} & 0 \\ 0 & \widetilde{f}_{\mathrm{vv}} \end{pmatrix}$$

The density matrix is unchanged

For a closed-shell determinant,

$$\widetilde{P} = 2\widetilde{C}_{\mathrm{o}} \cdot \widetilde{C}_{\mathrm{o}}^{\dagger} = 2C_{\mathrm{o}} \cdot C_{\mathrm{o}}^{\dagger} = P$$

The determinant changes only by a phase

$$|\widetilde{\Psi}_0\rangle = \det(U)\,|\Psi_0\rangle$$

The density, energy, and every expectation value are unchanged

- Only occupied-occupied or virtual-virtual rotations are invariant
- Mixing occupied and virtual orbitals changes the wavefunction and the energy
- Localization functionals are generally nonconvex: different guesses may converge to different local extrema

Localized orbitals

$$\boxed{\widetilde{\boldsymbol{C}}_{\mathrm{o}} = \boldsymbol{C}_{\mathrm{o}} \cdot \boldsymbol{U} \qquad \boldsymbol{U}^{\dagger} \cdot \boldsymbol{U} = \boldsymbol{1} \quad \Leftrightarrow \quad \text{unitary transformation}}$$

The density matrix is unchanged

For a closed-shell determinant,

$$\widetilde{\boldsymbol{P}} = 2\widetilde{\boldsymbol{C}}_{\mathrm{o}} \cdot \widetilde{\boldsymbol{C}}_{\mathrm{o}}^{\dagger} = 2\boldsymbol{C}_{\mathrm{o}} \cdot \boldsymbol{C}_{\mathrm{o}}^{\dagger} = \boldsymbol{P}$$

The determinant changes only by a phase

$$|\widetilde{\Psi}_0\rangle = \det(\boldsymbol{U})\,|\Psi_0\rangle$$

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Localization procedures

Edmiston-Ruedenberg functional

$$\boxed{\Omega_{\text{ER}} = \sum_i (\psi_i \psi_i | \psi_i \psi_i) \quad \longrightarrow \quad \text{max orbital self-repulsion}}$$

Foster-Boys functional

$$\begin{aligned}
\Omega_{\text{BF}} &= \frac{1}{2} \sum_i \left[\langle \psi_i \psi_i | r_{12}^2 | \psi_i \psi_i \rangle \right] \quad \longrightarrow \quad \text{min spatial extent} \\
&\equiv \sum_i \left[\langle \psi_i | r^2 | \psi_i \rangle - | \langle \psi_i | \boldsymbol{r} | \psi_i \rangle |^2 \right] \quad \longrightarrow \quad \text{max distance between centroids}
\end{aligned}$$

Strengths

- ✓ Requires only one-electron integrals
- ✓ Produces compact bonds and lone pairs

Limitations

- ✗ Does not explicitly preserve σ/π separation
- ✗ Diffuse virtual orbitals may be difficult to localize

Pipek-Mezey localization procedure

For an atomic population q_i^A of orbital i on atom A, the original Pipek-Mezey criterion maximizes

$$\boxed{\Omega_{\mathrm{PM}} = \sum_A \sum_i |q_i^A|^2}$$

using Mulliken populations

$$q_i^A = \sum_{\mu \in A} \sum_\nu C_{\mu i}^* S_{\mu\nu} C_{\nu i}$$

Strengths

- ✓ Often preserves separation between σ and π orbitals
- ✓ Produces intuitive bonds, lone pairs, and core orbitals

Limitations

- ✗ The original Mulliken form can be basis set dependent
- ✗ Löwdin populations or intrinsic atomic orbitals provide more robust alternatives

There is no universally *best* localized orbital set: the criterion should match the intended interpretation or computational use

Summary

What remains unchanged

- The occupied subspace and Slater determinant
- The density matrix and electron density
- The HF energy and molecular properties
- The zero occupied-virtual Fock block

$$\tilde{\boldsymbol{f}}_{\mathrm{ov}} = \boldsymbol{f}_{\mathrm{ov}} = \boldsymbol{0}$$

What changes

- Localized orbitals are generally not eigenfunctions of the Fock operator:

$$\tilde{\boldsymbol{f}}_{\mathrm{oo}} = \boldsymbol{U}^{\dagger} \cdot \boldsymbol{\epsilon}_{\mathrm{o}} \cdot \boldsymbol{U}$$

- Orbital energies cannot therefore be assigned to localized orbitals

Why localize?

- Identify bonds, lone pairs, and fragments
- Enable local correlation and embedding methods
- Select active spaces and domains
- Construct Wannier functions in periodic systems

Practical cautions

- Localize occupied and virtual spaces separately
- Core and valence orbitals are often localized separately
- Check for multiple extrema and discontinuous solutions
- Spatial symmetry may be lost

Unrestricted HF

How to model open-shell systems?

- For open-shell systems one uses unrestricted spin orbitals

$$\chi_p^{\text{UHF}}(\boldsymbol{x}) = \begin{cases} \alpha(\omega)\psi_p^{\alpha}(\boldsymbol{r}) \\ \beta(\omega)\psi_p^{\beta}(\boldsymbol{r}) \end{cases}$$

- UHF may break spin symmetry: the determinant need not be an eigenfunction of $\hat{S}^2$
- The HF wavefunction is no longer a pure spin function
- Singlets are contaminated by triplets, quintets, etc
- Doublets are contaminated by quartets, sextets, etc
- Generalized HF also allows spin mixing and need not preserve $\hat{S}_z$

$$\chi_p^{\text{GHF}}(\boldsymbol{x}) = \alpha(\omega)\psi_p^{\alpha}(\boldsymbol{r}) + \beta(\omega)\psi_p^{\beta}(\boldsymbol{r}) \quad \Rightarrow \text{two-component spinor}$$

UHF equations for unrestricted spin orbitals

To minimize the UHF energy, the unrestricted spatial orbitals must be eigenfunctions of the α and β Fock operators:

$$\boxed{f^{\alpha}(\boldsymbol{r}_1)\psi_i^{\alpha}(\boldsymbol{r}_1) = \epsilon_i^{\alpha}\psi_i^{\alpha}(\boldsymbol{r}_1)} \qquad \boxed{f^{\beta}(\boldsymbol{r}_1)\psi_i^{\beta}(\boldsymbol{r}_1) = \epsilon_i^{\beta}\psi_i^{\beta}(\boldsymbol{r}_1)}$$

where

$$f^{\alpha}(\boldsymbol{r}_1) = h(\boldsymbol{r}_1) + \sum_i^{N^{\alpha}}[J_i^{\alpha}(\boldsymbol{r}_1) - K_i^{\alpha}(\boldsymbol{r}_1)] + \sum_i^{N^{\beta}} J_i^{\beta}(\boldsymbol{r}_1)$$

$$f^{\beta}(\boldsymbol{r}_1) = h(\boldsymbol{r}_1) + \sum_i^{N^{\beta}}[J_i^{\beta}(\boldsymbol{r}_1) - K_i^{\beta}(\boldsymbol{r}_1)] + \sum_i^{N^{\alpha}} J_i^{\alpha}(\boldsymbol{r}_1)$$

The Coulomb and Exchange operators are

$$J_i^{\sigma}(\boldsymbol{r}_1) = \int \psi_i^{\sigma *}(\boldsymbol{r}_2) r_{12}^{-1} \psi_i^{\sigma}(\boldsymbol{r}_2)\, \mathrm{d}\boldsymbol{r}_2 \qquad K_i^{\sigma}(\boldsymbol{r}_1)\psi_j^{\sigma}(\boldsymbol{r}_1) = \left[\int \psi_i^{\sigma *}(\boldsymbol{r}_2) r_{12}^{-1} \psi_j^{\sigma}(\boldsymbol{r}_2)\, \mathrm{d}\boldsymbol{r}_2\right]\psi_i^{\sigma}(\boldsymbol{r}_1)$$

UHF energy

The UHF energy comprises three contributions:

$$E_{\text{UHF}} = E_{\text{UHF}}^{\alpha\alpha} + E_{\text{UHF}}^{\beta\beta} + E_{\text{UHF}}^{\alpha\beta}$$

which yields

$$\boxed{E_{\text{UHF}} = \sum_{i}^{N^\alpha} h_{ii}^{\alpha} + \frac{1}{2}\sum_{ij}^{N^\alpha}(J_{ij}^{\alpha\alpha} - K_{ij}^{\alpha\alpha}) + \sum_{i}^{N^\beta} h_{ii}^{\beta} + \frac{1}{2}\sum_{ij}^{N^\beta}(J_{ij}^{\beta\beta} - K_{ij}^{\beta\beta}) + \sum_{i}^{N^\alpha}\sum_{j}^{N^\beta} J_{ij}^{\alpha\beta}}$$

The matrix elements are given by

$$h_{ii}^{\sigma} = \left\langle \psi_i^{\sigma} \middle| h \middle| \psi_i^{\sigma} \right\rangle \qquad J_{ij}^{\sigma\sigma'} = \langle \psi_i^{\sigma} \psi_j^{\sigma'} | \psi_i^{\sigma} \psi_j^{\sigma'} \rangle \qquad K_{ij}^{\sigma\sigma} = \langle \psi_i^{\sigma} \psi_j^{\sigma} | \psi_j^{\sigma} \psi_i^{\sigma} \rangle$$

Note that there is no exchange between opposite-spin electrons

UHF energy of the Li atom

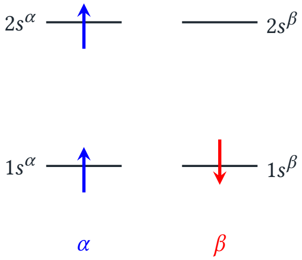
Problem

“Write down the UHF energy of the doublet state of the lithium atom”

UHF energy of the Li atom

Problem

"Write down the UHF energy of the doublet state of the lithium atom"



Solution

$$E_{\text{UHF}} = h_{11}^{\alpha} + h_{11}^{\beta} + h_{22}^{\alpha} + J_{12}^{\alpha\alpha} - K_{12}^{\alpha\alpha} + J_{11}^{\alpha\beta} + J_{21}^{\alpha\beta}$$

Expansion of the unrestricted spin orbitals in a basis

$$\psi_i^\alpha(\boldsymbol{r}) = \sum_{\mu=1}^{K} C_{\mu i}^\alpha \phi_\mu(\boldsymbol{r}) \qquad \psi_i^\beta(\boldsymbol{r}) = \sum_{\mu=1}^{K} C_{\mu i}^\beta \phi_\mu(\boldsymbol{r})$$

The Pople-Nesbet equations

$$\boldsymbol{F}^\alpha \cdot \boldsymbol{C}^\alpha = \boldsymbol{S} \cdot \boldsymbol{C}^\alpha \cdot \boldsymbol{E}^\alpha \qquad \boldsymbol{F}^\beta \cdot \boldsymbol{C}^\beta = \boldsymbol{S} \cdot \boldsymbol{C}^\beta \cdot \boldsymbol{E}^\beta$$

$$F_{\mu\nu}^\alpha = H_{\mu\nu} + \sum_{\lambda\sigma} P_{\lambda\sigma}^\alpha [(\mu\nu|\sigma\lambda) - (\mu\lambda|\sigma\nu)] + \sum_{\lambda\sigma} P_{\lambda\sigma}^\beta (\mu\nu|\sigma\lambda)$$
$$F_{\mu\nu}^\beta = H_{\mu\nu} + \sum_{\lambda\sigma} P_{\lambda\sigma}^\beta [(\mu\nu|\sigma\lambda) - (\mu\lambda|\sigma\nu)] + \sum_{\lambda\sigma} P_{\lambda\sigma}^\alpha (\mu\nu|\sigma\lambda)$$

$\boldsymbol{F}^\alpha$ and $\boldsymbol{F}^\beta$ are both functions of $\boldsymbol{C}^\alpha$ and $\boldsymbol{C}^\beta$ $\Rightarrow$ There's a coupling between α and β orbitals!

Unrestricted Density Matrices

Spin-up and spin-down density matrices

$$\boxed{P^{\alpha}_{\mu\nu} = \sum_{i=1}^{N^{\alpha}} C^{\alpha}_{\mu i} C^{\alpha *}_{\nu i} \quad \Leftrightarrow \quad \boldsymbol{P}^{\alpha}}$$

$$\boxed{P^{\beta}_{\mu\nu} = \sum_{i=1}^{N^{\beta}} C^{\beta}_{\mu i} C^{\beta *}_{\nu i} \quad \Leftrightarrow \quad \boldsymbol{P}^{\beta}}$$

Properties of the density ($\sigma = \alpha$ or β)

$$\rho^{\sigma}(\boldsymbol{r}) = \sum_{\mu\nu} \phi_{\mu}(\boldsymbol{r}) P^{\sigma}_{\mu\nu} \phi^{*}_{\nu}(\boldsymbol{r})$$

$$\int \rho^{\sigma}(\boldsymbol{r})\, \mathrm{d}\boldsymbol{r} = N^{\sigma}$$

Total and Spin density matrices

$$\underbrace{\boldsymbol{P}^{\mathrm{T}}}_{\text{Charge density}} = \boldsymbol{P}^{\alpha} + \boldsymbol{P}^{\beta}$$

$$\underbrace{\boldsymbol{P}^{\mathrm{S}}}_{\text{Spin density}} = \boldsymbol{P}^{\alpha} - \boldsymbol{P}^{\beta}$$

How to perform a UHF calculation in practice

The SCF algorithm

1. Specify molecule $\{\boldsymbol{R}_A\}$ and $\{Z_A\}$ and basis set $\{\phi_\mu\}$ (same as RHF)
2. Calculate integrals $S_{\mu\nu}$, $H_{\mu\nu}$ and $\langle\mu\nu|\lambda\sigma\rangle$ (same as RHF)
3. Diagonalize $\boldsymbol{S}$ and compute $\boldsymbol{X}$ (same as RHF)
4. Choose initial density matrices $\boldsymbol{P}^\alpha$ and $\boldsymbol{P}^\beta$
   - 1a. Calculate $\boldsymbol{G}^\alpha$ and then $\boldsymbol{F}^\alpha = \boldsymbol{H} + \boldsymbol{G}^\alpha$
   - 1b. Calculate $\boldsymbol{G}^\beta$ and then $\boldsymbol{F}^\beta = \boldsymbol{H} + \boldsymbol{G}^\beta$
   - 2. Compute $(\boldsymbol{F}^\alpha)' = \boldsymbol{X}^\dagger \cdot \boldsymbol{F}^\alpha \cdot \boldsymbol{X}$ and $(\boldsymbol{F}^\beta)' = \boldsymbol{X}^\dagger \cdot \boldsymbol{F}^\beta \cdot \boldsymbol{X}$
   - 3a. Diagonalize $(\boldsymbol{F}^\alpha)'$ to obtain $(\boldsymbol{C}^\alpha)'$ and $\boldsymbol{E}^\alpha$
   - 3b. Diagonalize $(\boldsymbol{F}^\beta)'$ to obtain $(\boldsymbol{C}^\beta)'$ and $\boldsymbol{E}^\beta$
   - 4. Calculate $\boldsymbol{C}^\alpha = \boldsymbol{X} \cdot (\boldsymbol{C}^\alpha)'$ and $\boldsymbol{C}^\beta = \boldsymbol{X} \cdot (\boldsymbol{C}^\beta)'$
   - 5. Form the new density matrices $\boldsymbol{P}^\alpha$ and $\boldsymbol{P}^\beta$, and compute $\boldsymbol{P}^\mathsf{T} = \boldsymbol{P}^\alpha + \boldsymbol{P}^\beta$
   - 6. Am I converged? If not go back to 1.
5. Evaluate the desired observables, including E_{UHF}

Hartree-Fock for minimal-basis H_2

Start from one normalized $1s$ function on each atom

$$\boxed{\psi_g = \frac{\phi_A + \phi_B}{\sqrt{2(1+S)}} \qquad \psi_u = \frac{\phi_A - \phi_B}{\sqrt{2(1-S)}}} \quad \text{with} \quad S = \langle \phi_A | \phi_B \rangle$$

Why is it special?

For two electrons in two spatial orbitals, UHF reduces to a single-parameter problem

RHF problem

RHF wavefunction and energy

$$|\Psi_{\mathrm{RHF}}\rangle = \left|\psi_g\bar{\psi}_g\right\rangle \quad \text{and} \quad E_{\mathrm{RHF}} = 2h_{gg} + J_{gg} + \frac{1}{R}$$

At large R, where $S \to 0$,

$$\psi_g(1)\psi_g(2) \longrightarrow \frac{1}{2}\left[\underbrace{\phi_A(1)\phi_B(2) + \phi_B(1)\phi_A(2)}_{\text{covalent}} + \underbrace{\phi_A(1)\phi_A(2) + \phi_B(1)\phi_B(2)}_{\text{ionic}}\right]$$

The RHF dissociation error

RHF retains 50% ionic character at $R \to \infty$. Both electrons are forced to share the same spatial orbital, so the RHF energy lies above the energy of two separated H atoms

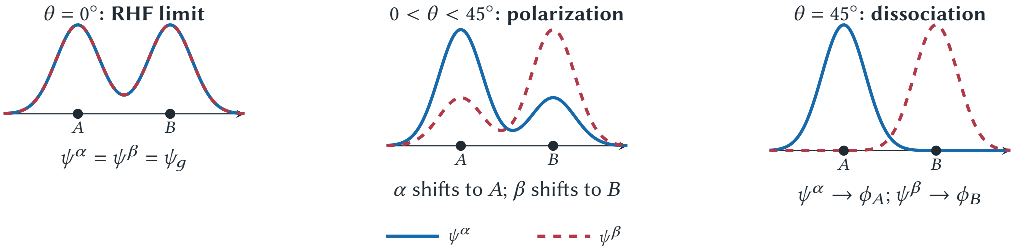
$\Rightarrow$ RHF is qualitatively correct near equilibrium but not at dissociation

UHF problem

UHF orbitals

$$\boxed{\begin{aligned}\psi^\alpha(\theta) &= \cos\theta\,\psi_g + \sin\theta\,\psi_u \\ \psi^\beta(\theta) &= \cos\theta\,\psi_g - \sin\theta\,\psi_u\end{aligned}} \quad \text{with} \quad \left\langle \psi^\alpha \middle| \psi^\beta \right\rangle = \cos(2\theta)$$

The two choices $\pm\theta$ are degenerate and exchange the spin localization



UHF energy and stationary points

With $J_{pq} = (pp|qq)$ and $K_{pq} = (pq|pq)$,

$$E_{\text{UHF}}(\theta) = 2\cos^2\theta\, h_{gg} + 2\sin^2\theta\, h_{uu} + \cos^4\theta\, J_{gg} + \sin^4\theta\, J_{uu} + 2\sin^2\theta\cos^2\theta\big(J_{gu} - 2K_{gu}\big)$$

Besides the restricted stationary point $\theta = 0$, a broken-symmetry stationary solution satisfies

$$\boxed{\cos^2\theta^\star = \frac{h_{uu} - h_{gg} + J_{uu} - J_{gu} + 2K_{gu}}{J_{gg} + J_{uu} - 2J_{gu} + 4K_{gu}}}$$

Short bond length

Only the minimum at $\theta = 0$ is physically accessible, and UHF reduces to RHF

Large bond length

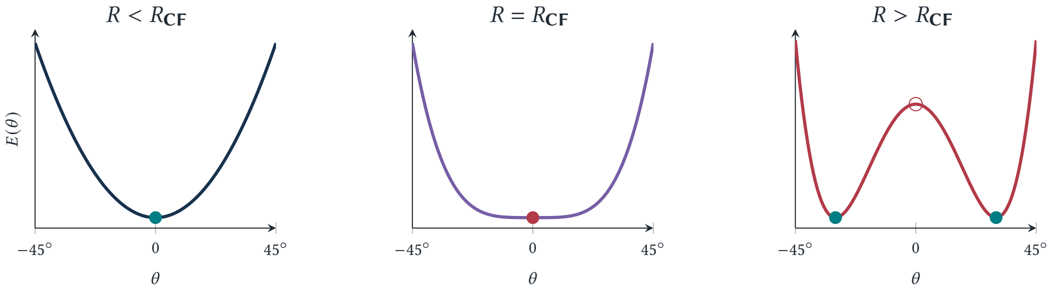
When $0 \leq \cos^2\theta^\star \leq 1$, two minima appear at $\pm\theta^\star$, and RHF becomes unstable to spin polarization

Coulson-Fischer (CF) point

$$\left.\frac{\mathrm{d}^2E}{\mathrm{d}\theta^2}\right|_{\theta=0} = 4\left(h_{uu} - h_{gg} - J_{gg} + J_{gu} - 2K_{gu}\right)$$

$$\boxed{R = R_{\text{CF}} \quad\Longleftrightarrow\quad \left.\frac{\mathrm{d}^2E}{\mathrm{d}\theta^2}\right|_{\theta=0} = 0}$$

The Coulson-Fischer point

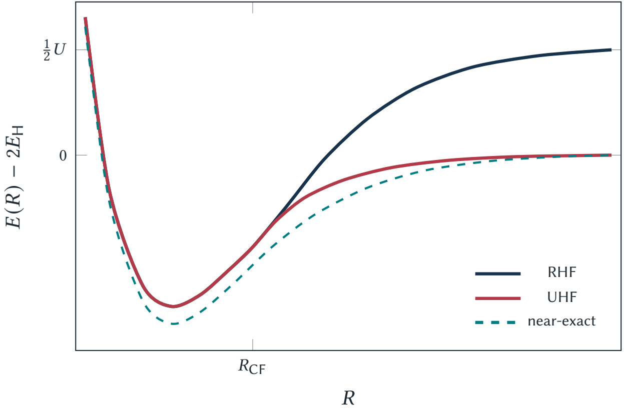


Positive RHF curvature	Vanishing RHF curvature	Negative RHF curvature
one minimum at $\theta = 0$	the minimum becomes flat	two UHF minima at $\pm\theta^\star$

What changes at the Coulson-Fischer point?

The RHF determinant at $\theta = 0$ does not disappear: it changes from a minimum to an unstable stationary point while two degenerate UHF minima emerge at $\pm\theta^\star$

Schematic dissociation curve



At $R \longrightarrow \infty$,

$$S \longrightarrow 0 \qquad \psi_g \longrightarrow \frac{\phi_A + \phi_B}{\sqrt{2}}$$

Define the positive on-site repulsion

$$U = (AA|AA)$$

Restricted limit

$$\boxed{E_{\mathrm{RHF}}^{\infty} = 2E_{\mathrm{H}} + \frac{1}{2}U}$$

The ionic terms leave a finite error

Unrestricted limit

Since $\psi^{\alpha} \longrightarrow \phi_A$ and $\psi^{\beta} \longrightarrow \phi_B$,

$$\boxed{E_{\mathrm{UHF}}^{\infty} = 2E_{\mathrm{H}}}$$

Symmetry-broken UHF

For the two-electron $M_S = 0$ determinant,

$$\boxed{\left\langle \hat{S}^2 \right\rangle_{\text{UHF}} = 1 - \left| \left\langle \psi^\alpha \middle| \psi^\beta \right\rangle \right|^2 = \sin^2(2\theta)}$$

Near equilibrium

$\theta = 0$ and $\langle \hat{S}^2 \rangle = 0$: the determinant is a pure singlet

At dissociation

$\theta \to \pi/4$ and $\langle \hat{S}^2 \rangle \to 1$: the determinant is an equal mixture of singlet and $M_S = 0$ triplet

$$|A\alpha\, B\beta\rangle = \frac{1}{\sqrt{2}} \left(|\text{covalent singlet}\rangle + |\text{triplet}, M_S = 0\rangle \right) \quad \text{as} \quad R \to \infty$$

$\Longrightarrow$ The UHF energy is physically useful, but the UHF wavefunction is spin-contaminated

Three take-home messages

1. A stationary SCF solution need not be stable
2. The Coulson-Fischer point marks a continuous RHF to UHF transition
3. Symmetry breaking mimics static correlation at the price of a contaminated wavefunction

Generalized Hartree-Fock

Key idea: GHF allows every orbital to choose its spin direction

A GHF orbital is a two-component spinor:

$$\boxed{\chi_p^{\text{GHF}}(\boldsymbol{x}) = \alpha(\omega)\,\psi_p^{\alpha}(\boldsymbol{r}) + \beta(\omega)\,\psi_p^{\beta}(\boldsymbol{r})}$$

Both components are expanded in the same spatial AO basis

$$\psi_i^{\sigma}(\boldsymbol{r}) = \sum_{\mu=1}^{K} C_{\mu i}^{\sigma}\phi_{\mu}(\boldsymbol{r}) \quad \text{with} \quad \boldsymbol{C} = \begin{pmatrix} \boldsymbol{C}^{\alpha} \\ \boldsymbol{C}^{\beta} \end{pmatrix} \quad \text{and} \quad \boldsymbol{C} \in \mathbb{C}^{2K\times N}$$

RHF

Paired electrons share one spatial orbital: both $\hat{S}^2$ and $\hat{S}_z$ are preserved

UHF

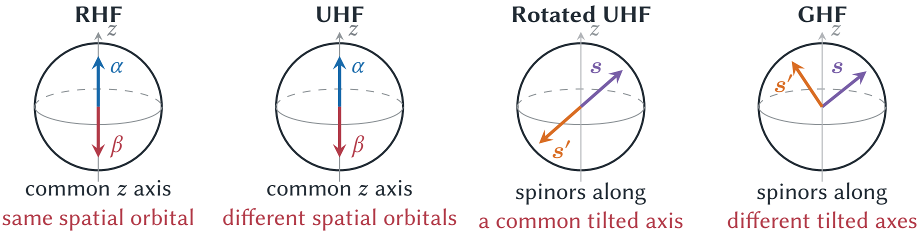
Each orbital is purely α or β: the spin density is collinear and $\hat{S}_z$ is preserved

GHF

Each orbital may mix α and β: in general, neither $\hat{S}^2$ nor $\hat{S}_z$ is preserved

$\Longrightarrow$ GHF does not require a common spin-quantization axis or fixed values of N^{α} and N^{β}

GHF allows arbitrary spin directions



Collinear vs non-collinear solutions

$$\underbrace{\text{UHF} \xrightarrow{\text{global spin rotation}} \text{rotated UHF}}_{\text{physically equivalent collinear solutions}} \quad \neq \quad \underbrace{\text{genuine GHF}}_{\text{non-collinear}} \tag{2}$$

$\Rightarrow$ A tilted spin axis alone does not imply a genuinely non-collinear GHF solution

GHF density matrix

Density matrix

$$\boxed{\mathcal{P} = \begin{pmatrix} \boldsymbol{P}^{\alpha\alpha} & \boldsymbol{P}^{\alpha\beta} \\ \boldsymbol{P}^{\beta\alpha} & \boldsymbol{P}^{\beta\beta} \end{pmatrix}} \quad \text{with} \quad P_{\mu\nu}^{\sigma\sigma'} = \sum_{i=1}^{N} C_{\mu i}^{\sigma} C_{\nu i}^{\sigma' *}$$

What do the blocks mean?

- $\boldsymbol{P} = \frac{1}{2}\left(\boldsymbol{P}^{\alpha\alpha} + \boldsymbol{P}^{\beta\beta}\right)$ determines the charge density
- $\boldsymbol{M}_z = \frac{1}{2}\left(\boldsymbol{P}^{\alpha\alpha} - \boldsymbol{P}^{\beta\beta}\right)$ gives the z component of magnetization
- $\boldsymbol{M}_y = \frac{1}{2\mathrm{i}}\left(\boldsymbol{P}^{\beta\alpha} - \boldsymbol{P}^{\alpha\beta}\right)$ generates the transverse y component
- $\boldsymbol{M}_x = \frac{1}{2}\left(\boldsymbol{P}^{\beta\alpha} + \boldsymbol{P}^{\alpha\beta}\right)$ generates the transverse x component

$$\mathcal{P} = \begin{pmatrix} \boldsymbol{P} + \boldsymbol{M}_z & \boldsymbol{M}_x - \mathrm{i}\boldsymbol{M}_y \\ \boldsymbol{M}_x + \mathrm{i}\boldsymbol{M}_y & \boldsymbol{P} - \boldsymbol{M}_z \end{pmatrix} = \boldsymbol{P} \otimes \mathbf{1} + \boldsymbol{M} \otimes \boldsymbol{\sigma} \quad \text{with} \quad \boldsymbol{M} = \begin{pmatrix} \boldsymbol{M}_x \\ \boldsymbol{M}_y \\ \boldsymbol{M}_z \end{pmatrix} \quad \text{and} \quad \boldsymbol{\sigma} = \begin{pmatrix} \boldsymbol{\sigma}_x \\ \boldsymbol{\sigma}_y \\ \boldsymbol{\sigma}_z \end{pmatrix}$$

GHF equations: Roothaan-Hall in spinor space

Generalized eigenvalue problem

$$\boxed{\boldsymbol{\mathcal{F}} \cdot \boldsymbol{\mathcal{C}} = \boldsymbol{\mathcal{S}} \cdot \boldsymbol{\mathcal{C}} \cdot \boldsymbol{\mathcal{E}}} \quad \text{with} \quad \boldsymbol{\mathcal{S}} = \begin{pmatrix} \boldsymbol{S} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{S} \end{pmatrix} \quad \text{and} \quad \boldsymbol{\mathcal{C}}^\dagger \cdot \boldsymbol{\mathcal{S}} \cdot \boldsymbol{\mathcal{C}} = \boldsymbol{1}$$

Fock matrix

$$\boxed{\boldsymbol{\mathcal{F}} = \begin{pmatrix} \boldsymbol{h} + \boldsymbol{J}[\boldsymbol{P}^{\alpha\alpha}] + \boldsymbol{J}[\boldsymbol{P}^{\beta\beta}] - \boldsymbol{K}[\boldsymbol{P}^{\alpha\alpha}] & -\boldsymbol{K}[\boldsymbol{P}^{\alpha\beta}] \\ -\boldsymbol{K}[\boldsymbol{P}^{\beta\alpha}] & \boldsymbol{h} + \boldsymbol{J}[\boldsymbol{P}^{\alpha\alpha}] + \boldsymbol{J}[\boldsymbol{P}^{\beta\beta}] - \boldsymbol{K}[\boldsymbol{P}^{\beta\beta}] \end{pmatrix}}$$

with

$$J_{\mu\nu}[\boldsymbol{P}] = \sum_{\lambda\sigma} P_{\lambda\sigma} (\mu\nu|\lambda\sigma) \qquad K_{\mu\nu}[\boldsymbol{P}] = \sum_{\lambda\sigma} P_{\lambda\sigma} (\mu\sigma|\lambda\nu)$$

The off-diagonal exchange blocks couple the α and β components self-consistently

SCF procedure

Build all four density blocks $\longrightarrow$ build all four Fock blocks $\longrightarrow$ solve a $2K \times 2K$ generalized eigenvalue problem $\longrightarrow$ occupy the N lowest spinors $\longrightarrow$ iterate

RHF, UHF, and GHF

$$\boxed{\{\Phi_{\mathrm{RHF}}\} \subset \{\Phi_{\mathrm{UHF}}\} \subset \{\Phi_{\mathrm{GHF}}\}} \quad \Longrightarrow \quad E_{\mathrm{GHF}} \le E_{\mathrm{UHF}} \le E_{\mathrm{RHF}}$$

for the corresponding global minima in the same one-electron basis

A new stability direction

A converged UHF determinant may still be unstable to occupied-virtual rotations that mix α and β orbitals. Following such a negative-curvature direction produces a lower-energy GHF solution.

Not every GHF solution is noncollinear

RHF and UHF are special cases of GHF. A global spin rotation of a UHF determinant changes its apparent quantization axis but remains physically collinear.

Symmetry dilemma

Lowering the variational energy may require breaking exact symmetries:

- UHF may break $\hat{\mathcal{S}}^2$
- GHF may break both $\hat{\mathcal{S}}^2$ and $\hat{\mathcal{S}}_z$
- Complex GHF may also break time-reversal symmetry

Some comments on GHF

Interpret with care

- GHF is still a single determinant: correlation is not included
- S and M_S are generally not good quantum numbers
- Many stationary solutions make the initial guess important: stability analysis
- The GHF solution might be a globally rotated UHF solution: collinearity test
- Equilateral H_3 and triangular magnetic clusters are typical examples of genuine noncollinearity

Take-home messages

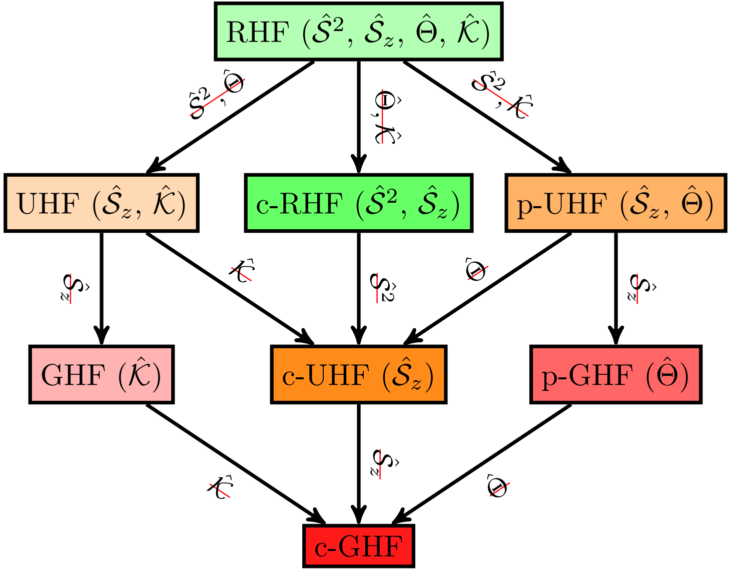
- GHF is the most general number-conserving single-determinant HF ansatz
- One can break particle-number symmetry within Hartree-Fock-Bogoliubov (HFB) theory

Stuber-Paldus classification of Hartree-Fock solutions

Fukutome designation	Stuber-Paldus designation	Symmetries preserved[i]	Structure of the orbital coefficient matrix C
TICS[a]	real RHF	$\hat{\mathcal{S}}^2$, $\hat{\mathcal{S}}_z$, $\hat{\mathcal{K}}$, $\hat{\Theta}$	$\begin{pmatrix} C_{\sigma\sigma} & \mathbf{0} \\ \mathbf{0} & C_{\sigma\sigma} \end{pmatrix}, \quad C \in \mathbb{R}$
CCW[b]	complex RHF	$\hat{\mathcal{S}}^2$, $\hat{\mathcal{S}}_z$	$\begin{pmatrix} C_{\sigma\sigma} & \mathbf{0} \\ \mathbf{0} & C_{\sigma\sigma} \end{pmatrix}$
ASCW[c]	paired UHF	$\hat{\mathcal{S}}_z$, $\hat{\Theta}$	$\begin{pmatrix} C_{\sigma\sigma} & \mathbf{0} \\ \mathbf{0} & C^*_{\sigma\sigma} \end{pmatrix}$
ASDW[d]	real UHF	$\hat{\mathcal{S}}_z$, $\hat{\mathcal{K}}$	$\begin{pmatrix} C_{\sigma\sigma} & \mathbf{0} \\ \mathbf{0} & C_{\sigma'\sigma'} \end{pmatrix}, \quad C \in \mathbb{R}$
ASW[e]	complex UHF	$\hat{\mathcal{S}}_z$	$\begin{pmatrix} C_{\sigma\sigma} & \mathbf{0} \\ \mathbf{0} & C_{\sigma'\sigma'} \end{pmatrix}$
TSCW[f]	paired GHF	$\hat{\Theta}$	$\begin{pmatrix} C_{\sigma\sigma} & C_{\sigma\sigma'} \\ -C^*_{\sigma\sigma'} & C^*_{\sigma\sigma} \end{pmatrix}$
TSDW[g]	real GHF	$\hat{\mathcal{K}}$	$\begin{pmatrix} C_{\sigma\sigma} & C_{\sigma\sigma'} \\ C_{\sigma'\sigma} & C_{\sigma'\sigma'} \end{pmatrix}, \quad C \in \mathbb{R}$
TSW[h]	complex GHF		$\begin{pmatrix} C_{\sigma\sigma} & C_{\sigma\sigma'} \\ C_{\sigma'\sigma} & C_{\sigma'\sigma'} \end{pmatrix}$

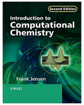
[a] Time-reversal invariant closed-shell. [b] Charge current wave. [c] Axial spin current wave. [d] Axial spin density wave. [e] Axial spin wave. [f] Torsional spin current wave. [g] Torsional spin density wave. [h] Torsional spin wave. [i] $\hat{\mathcal{K}}$ and $\hat{\Theta}$ are the complex conjugation and time reversal operators.

Stuber-Paldus classification of Hartree-Fock solutions

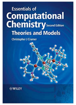


Recommended references

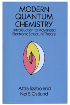
- Introduction to Computational Chemistry (Jensen)



- Essentials of Computational Chemistry (Cramer)



- Modern Quantum Chemistry (Szabo & Ostlund)



- Molecular Electronic Structure Theory (Helgaker, Jørgensen & Olsen)